Quantum Computing
Professor Gopal Pandurangan

Scribe Author: Armaan Amatya

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Week: 1 (August 24–28, 2026)

Lecture Date: Tuesday, August 25, 2026

1 Efficient Algorithms: Primes and Factoring

The input of the primality problem is the binary representation of n, of size $N = \log_2 n$ bits. An algorithm is **efficient** if it runs in time polynomial in N.

Trial division takes $\Theta(\sqrt{n}) = \Theta\left(2^{N/2}\right)$ time, which is exponential. A deterministic polynomial-time algorithm exists ($\tilde{\mathcal{O}}(N^6)$) but is not used in practice. The *randomized* Miller–Rabin test takes $\mathcal{O}(N^3)$ time: a composite n has at least $3/4$ of $\{2, \ldots, n-2\}$ as *witnesses* and a prime has none, so k independent samples miss a composite with probability at most $(1/4)^k$.

Factoring. Given $n = pq$ with p, q prime, find p and q. Factoring is an *encryption primitive*. No polynomial-time classical algorithm is known, but *Shor's factoring algorithm* is a quantum algorithm with running time $\mathcal{O}(N^3)$. No quantum algorithm is known that solves NP-complete problems such as 3-SAT in polynomial time.

2 Random Sampling

A randomized algorithm only needs to *sample*, and a (fair) coin flip is an $\mathcal{O}(1)$ time primitive. In an array with n slots (Figure 1), n a power of 2, every index has $\log n$ bits. One coin flip per bit samples a uniformly random element in $\mathcal{O}(\log n)$ time, since $\mathbb{P}(\text{index } i \text{ is chosen}) = (1/2)^{\log n} = 1/n$.

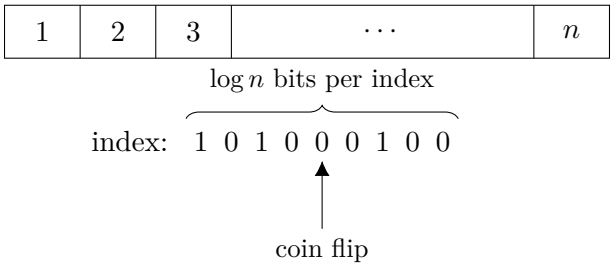


Figure 1: Each of the $\log n$ bits of the index is one coin flip.

We sample *with* replacement, so that the samples are independent. A sample hits a given element with probability $1/n$, so finding it takes n samples in expectation (Theorem 5).

3 Search

An unsorted array A with n slots is accessed by *queries* ("what is $A[i]$?"), and we want the index of a given element x. Linear search makes $\mathcal{O}(n)$ queries. The lower bound uses an *adversary argument*: the adversary builds an input on which the algorithm does badly.

Theorem 1 ▶ Lower Bound for Deterministic Search

Every deterministic algorithm for search makes at least $n - 1$ queries in the worst case.

Proof. Whenever the algorithm queries a slot i, the adversary answers that $A[i] \neq x$. The algorithm is deterministic, so this defines one run. Suppose that it stops after $k \leqslant n-2$ queries and outputs an index j. At least 2 slots have not been queried, so the adversary can place x in an unqueried slot different from j. This array is consistent with all the answers, and the algorithm is wrong on it. So a correct algorithm makes at least $n-1$ queries. □

Randomized search needs $\Omega(n)$ expected queries (*Yao's min-max principle*), and Grover's quantum algorithm needs $\mathcal{O}(\sqrt{n})$ queries. A *quantum advantage* is a problem with $\Omega(f(n))$ queries classically and $o(f(n))$ queries quantumly.

4 Probability

- **Sample space:** a finite set S of *simple events*, $|S| = n$.
- **Probability function:** $\mathbb{P}\colon S \to [0,1]$.
- **Event:** a subset $A \subseteq S$, with $\mathbb{P}(A) = \sum_{a \in A} \mathbb{P}(a)$.
- **Axioms:** $\mathbb{P}(A) \geqslant 0$, $\mathbb{P}(S) = 1$, and $\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \mathbb{P}(A_i)$ for pairwise disjoint events.
- **Random variable:** a function $X\colon S \to V$, with $\mathbb{P}(X = r) = \sum_{s \in S\,:\,X(s)=r} \mathbb{P}(s)$.

Week: 1 (August 24–28, 2026)

Lecture Date: Thursday, August 27, 2026

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1 Search and Satisfiability

- **Query** (in Grover's algorithm): returns the element at one index of the input.
- **SAT:** find an assignment with $f(x_1, \ldots, x_n) = 1$ for a boolean function f. No polynomial-time algorithm is known.
- **3-SAT:** $f = C_1 \wedge C_2 \wedge \cdots \wedge C_m$, at most three variables per clause, for example $C_1 = (x_2 \vee \bar{x}_3)$. A randomized algorithm takes $\mathcal{O}((4/3)^n)$ time.
- **Grover:** $\mathcal{O}\left(\sqrt{2^n}\right) = \mathcal{O}\left(2^{n/2}\right)$ queries ($\sqrt{2} \approx 1.41$), and the same for every problem that is a search for an input with a given function value.
- A quantum advantage needs lower bounds for *randomized* algorithms, not only for deterministic ones.

2 Random Sampling and Expectation

Random sampling is the basis of many randomized algorithms. Unless another distribution is given, the samples are uniform, so one sample from an array of n elements hits a fixed element with probability $1/n$.

Let S be a sample space with probability function $\mathbb{P}$ as in the lecture of August 25, except that S may now be countably infinite. For random sampling we take X to be the number of

queries until the element is found. Its values are $1, 2, 3, \ldots$ when we sample with replacement and $1, \ldots, n$ when we sample without replacement.

Definition 1 ▶ Expectation

The **expectation** of a random variable X, also called its *first moment*, is

$$\mathbb{E}[X] = \sum_{x \in \text{range}(X)} x \cdot \mathbb{P}(X = x).$$

All random variables in this lecture are nonnegative, so every sum below has nonnegative terms and can be reordered.

Theorem 2 ▶ Linearity of Expectation

Let $X = \sum_{i=1}^{n} X_i$, where the X_i are nonnegative random variables on the same sample space. Then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

Proof. Grouping the simple events s by the value $X(s)$ gives $\mathbb{E}[X] = \sum_{s \in S} X(s)\,\mathbb{P}(s)$. So

$$\mathbb{E}[X] = \sum_{s \in S} \sum_{i=1}^{n} X_i(s)\,\mathbb{P}(s) = \sum_{i=1}^{n} \sum_{s \in S} X_i(s)\,\mathbb{P}(s) = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

□

The X_i do not have to be independent.

Independence. Two events A and B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$, and **mutually exclusive** if $A \cap B = \emptyset$. The two notions are not the same. Two random variables X and Y are independent if $\mathbb{P}(X = x \text{ and } Y = y) = \mathbb{P}(X = x)\,\mathbb{P}(Y = y)$ for all x and y. Random variables $X_1, \ldots, X_n$ are independent if $\mathbb{P}(X_1 = x_1 \text{ and } \ldots \text{ and } X_n = x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i = x_i)$ for all values $x_1, \ldots, x_n$.

3 Bernoulli, Binomial and Geometric Random Variables

Three random variables appear again and again in the analysis of randomized algorithms.

Definition 2 ▶ Bernoulli Random Variable

A random variable is **Bernoulli** (or 0-1, or an *indicator*) with parameter p if it is 1 with probability p and 0 with probability $1 - p$.

Its expectation is $1 \cdot p + 0 \cdot (1 - p) = p$.

Definition 3 ▶ Binomial Random Variable

A **binomial** random variable is a sum of n independent Bernoulli random variables with the same parameter p,

$$B(n, p) = \sum_{i=1}^{n} X_i.$$

It counts the successes among n independent trials with success probability p.

A fixed pattern with k successes and $n-k$ failures has probability $p^k(1-p)^{n-k}$ by independence, and there are $\binom{n}{k}$ such patterns, so $\mathbb{P}(B(n,p)=k) = \binom{n}{k}p^k(1-p)^{n-k}$.

Theorem 3

$\mathbb{E}[B(n,p)] = \sum_{k=0}^{n} k \cdot \mathbb{P}(B(n,p)=k) = np.$

Proof. The first equality is Definition 1. By linearity of expectation (Theorem 2),

$$\mathbb{E}[B(n,p)] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} p = np.$$

□

Definition 4 ▶ Geometric Random Variable

Consider a sequence of independent Bernoulli trials that each succeed with probability $p > 0$. The number of trials up to and including the first success is a **geometric** random variable, $X \sim G(p)$.

The first success is in trial k if the first $k-1$ trials fail and trial k succeeds, so $\mathbb{P}(G(p)=k) = (1-p)^{k-1}p$ and

$$\mathbb{E}[G(p)] = \sum_{k=1}^{\infty} k\,(1-p)^{k-1}p.$$

We compute this sum with conditional expectation.

4 Conditional Probability and Conditional Expectation

Definition 5 ▶ Conditional Probability

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This is the probability of A once B is known to have happened.

Definition 6 ▶ Conditional Expectation

For random variables X and Y and a value y with $\mathbb{P}(Y=y) > 0$,

$$\mathbb{E}[X \mid Y=y] = \sum_{x \in \text{range}(X)} x \cdot \mathbb{P}(X=x \mid Y=y).$$

The **conditional expectation** $\mathbb{E}[X \mid Y]$ takes the value $\mathbb{E}[X \mid Y=y]$ when $Y=y$.

An expectation is a number, but $\mathbb{E}[X \mid Y]$ is a random variable. It is a function $f(Y)$ of Y.

Theorem 4

$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$

Proof. Let $f(y) = \mathbb{E}[X \mid Y = y]$. Grouping the simple events by the value of Y, as in the proof of Theorem 2, gives $\mathbb{E}[f(Y)] = \sum_y f(y)\,\mathbb{P}(Y = y)$, where the sums over y are over the values with $\mathbb{P}(Y = y) > 0$. By Definitions 5 and 6,

$$\begin{aligned}\mathbb{E}[\mathbb{E}[X \mid Y]] &= \sum_y \mathbb{P}(Y = y) \sum_x x \cdot \mathbb{P}(X = x \mid Y = y) = \sum_x x \sum_y \mathbb{P}(X = x \text{ and } Y = y) \\ &= \sum_x x \cdot \mathbb{P}(X = x) = \mathbb{E}[X],\end{aligned}$$

because the events $Y = y$ are mutually exclusive, one of them occurs, and the values y with $\mathbb{P}(Y = y) = 0$ add nothing to the sum. □

Theorem 5 ▶ Expectation of a Geometric Random Variable

$\mathbb{E}[G(p)] = 1/p$.

Proof. Let $X \sim G(p)$. If $p = 1$, then $X = 1$ always and $\mathbb{E}[X] = 1$. Let $p < 1$, and let $X_1 = 1$ if the first trial succeeds and $X_1 = 0$ otherwise. We condition on X_1. If $X_1 = 1$, then $X = 1$. If $X_1 = 0$, then $X \geqslant 2$, and for $k \geqslant 2$,

$$\mathbb{P}(X = k \mid X_1 = 0) = \frac{(1-p)^{k-1}p}{1-p} = (1-p)^{k-2}p = \mathbb{P}(X = k-1).$$

So $\mathbb{E}[X \mid X_1 = 0] = \sum_{j \geqslant 1}(j+1)\,\mathbb{P}(X = j) = \mathbb{E}[X] + 1$. By Theorem 4,

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[X \mid X_1 = 1]\,\mathbb{P}(X_1 = 1) + \mathbb{E}[X \mid X_1 = 0]\,\mathbb{P}(X_1 = 0) \\ &= p + (1-p)(\mathbb{E}[X] + 1) = (1-p)\,\mathbb{E}[X] + 1.\end{aligned}$$

The sum for $\mathbb{E}[X]$ converges by the ratio test, since consecutive terms have ratio $\frac{(k+1)(1-p)}{k} \to 1-p < 1$. So $\mathbb{E}[X]$ is finite and we can subtract $(1-p)\,\mathbb{E}[X]$ on both sides. This gives $p\,\mathbb{E}[X] = 1$ and $\mathbb{E}[X] = 1/p$. □

For random sampling with replacement, $X \sim G(1/n)$, so $\mathbb{E}[X] = n$.

Two classes of randomized algorithms. Randomized algorithms are either *Las Vegas* or *Monte Carlo* algorithms (Definition 7). A lower bound for randomized search (Theorem 8) is shown with Yao's min-max principle.

Week: 2 (August 31 – September 4, 2026)

Lecture Date: Tuesday, September 1, 2026

1 Query Complexity for Search

In the search problem of Figure 1, a *query* to a slot x of an array of n slots returns $f(x)$, and the goal is to find the slot x with $f(x) = 1$. The *query complexity* of an algorithm is the number of queries that it makes.

Definition 7 ▶ Monte Carlo and Las Vegas Algorithms

A **Monte Carlo** algorithm is a randomized algorithm whose error probability is > 0.
A **Las Vegas** algorithm is a randomized algorithm whose error probability is $= 0$.

Random sampling *with replacement* queries a slot chosen uniformly at random in every step. Every query succeeds with probability $1/n$, so the number of queries until the first success is a *geometric* random variable with parameter $1/n$, and $\mathbb{E}[\text{queries until find}] = n$ (Theorem 5). Stopped after a fixed number of queries, random sampling is a Monte Carlo algorithm, because it can fail to find the slot. Run until the slot is found, it is a Las Vegas algorithm with n expected queries.

2 Coupon Collector Problem

To distinguish 2 binary arrays with random sampling, all the n slots have to be accessed. The number of queries that this takes is the *coupon collector problem.*

Theorem 6 ▶ Coupon Collector

Let X be the number of queries made by random sampling (with replacement) until all the n slots are accessed. Then

$$\mathbb{E}[X] = nH_n = \Theta(n \log n),$$

where $H_n = \sum_{i=1}^{n} \frac{1}{i}$ is the nth harmonic number.

Proof. Write $X = X_1 + X_2 + \cdots + X_n$, where X_i is the number of queries made to access a new slot after $i-1$ different slots have been accessed. Such a query accesses a new slot with probability $\frac{n-(i-1)}{n}$. So $X_i \sim G\left(\frac{n-(i-1)}{n}\right)$ is geometric with $\mathbb{E}[X_i] = \frac{n}{n-(i-1)}$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n \sum_{i=1}^{n} \frac{1}{n-i+1} = nH_n,$$

and $H_n = \Theta(\log n)$. □

2.1 High Probability Bound

Here log is the natural logarithm.

Theorem 7

If random sampling makes $3n \log n$ queries, then

$$\mathbb{P}(\text{fail}) \leqslant \frac{1}{n^2}, \qquad \text{that is,} \qquad \mathbb{P}(\text{Success}) \geqslant 1 - \frac{1}{n^2},$$

where "fail" means that some slot is not accessed.

Proof. Let B_i be the event that slot i is not accessed. A query misses slot i with probability $1 - \frac{1}{n}$, and the queries are independent. Using $1 - x \leqslant e^{-x}$, we get

$$\mathbb{P}(B_i) \leqslant \left(1 - \frac{1}{n}\right)^{3n \log n} \leqslant e^{-3 \log n} = \frac{1}{n^3}, \qquad \mathbb{P}\left(\bigcup_i B_i\right) \leqslant \sum_{i=1}^{n} \mathbb{P}(B_i) \leqslant \frac{n}{n^3} = \frac{1}{n^2}$$

by the union bound. □

3 Lower Bound for Randomized Algorithms

Theorem 8

Any Las Vegas randomized algorithm for search needs $\Omega(n)$ expected queries.

Yao's Min-Max Principle

Let A^{rand} be a randomized algorithm, and let $\mathbb{E}\left[A^{\text{rand}}\right]$ be its expected number of queries on its worst input. Let μ be a probability distribution over the inputs, and let D_μ^{best} be the best deterministic algorithm for inputs chosen according to μ. Then

$$\mathbb{E}\left[A^{\text{rand}}\right] \geqslant \max_\mu \mathbb{E}_\mu\left[D_\mu^{\text{best}}\right] \geqslant \Omega(n).$$

The quantity $\mathbb{E}_\mu\left[D_\mu^{\text{best}}\right]$ is called the *distributional complexity*. The first inequality is the principle. The second one holds for search and proves Theorem 8.

Proof of Theorem 8. Let μ choose the slot x with $f(x) = 1$ uniformly at random among the n slots, and let D be a deterministic algorithm that is always correct. As long as all the answers are 0, D queries a fixed sequence of different slots $s_1, s_2, \ldots$, because repeating a query gives no new information. It cannot stop while all the answers are 0 and two slots are not yet queried: either of them could be the solution, and D would be wrong on one of the two inputs. So if the solution is s_j, then D makes at least $\min(j, n-1)$ queries. Every j has probability $1/n$, so

$$\mathbb{E}_\mu[D] \geqslant \frac{1}{n} \sum_{j=1}^{n} \min(j, n-1) \geqslant \frac{1}{n} \sum_{j=1}^{n-1} j = \frac{n-1}{2} = \Omega(n).$$ □

4 Chernoff Bounds

If a coin gives heads with probability $\frac{1}{2} + \varepsilon$ and is tossed n times, then $\mathbb{E}[\text{Heads}] = \frac{n}{2} + n\varepsilon$. Chernoff bounds show that the number of heads is close to its expectation, so $\mathbb{P}(\text{Heads} < \frac{n}{2})$ is very small.

In general, let

$$X = \sum_{i=1}^{n} X_i, \qquad X_i = \begin{cases} 1 & \text{with probability } p_i, \\ 0 & \text{with probability } 1 - p_i, \end{cases}$$

where the X_i are independent Bernoulli random variables. If $p_i = p$ for all i, then $\mathbb{E}[X] = np$ and $\mathbb{V}[X] = np(1-p)$. Figure 2 shows the distribution of X around its mean np.

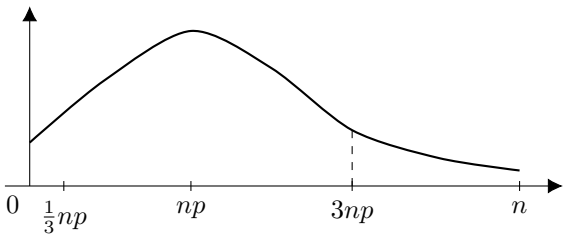


Figure 2: The distribution of X, with mean np.

Theorem 9 ▶ Chernoff Bounds

Let $X = \sum_{i=1}^{n} X_i$ be as above and let $\mu = \mathbb{E}[X]$.

- Upper tail (UI): for $0 < \delta < 1$,

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{3}}.$$

- For $R \geqslant 6\mu$,

$$\mathbb{P}(X \geqslant R) \leqslant \frac{1}{2^R}.$$

- Lower tail (LI): for $0 < \delta < 1$,

$$\mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{2}}.$$

4.1 Solution of the Exercise

Exercise 1. Show that if random sampling makes $24n \log n$ queries, then

$$\mathbb{P}(\text{some slot is accessed} < 12 \log n \text{ times}) \leqslant \frac{1}{n^2},$$

so for $n \geqslant 2$ and $c = 6$,

$$\mathbb{P}(\text{some slot is accessed} \leqslant c \log n \text{ times}) \leqslant \frac{1}{n^2}.$$

Proof. Fix a slot i and let $T = 24n \log n$ be the number of queries. Let $Y_i = \sum_{t=1}^{T} Z_t$ be the number of times that slot i is accessed, where $Z_t = 1$ if the tth query accesses slot i and $Z_t = 0$ otherwise. Sampling is with replacement, so the queries are independent, and each accesses slot i with probability $1/n$. So the Z_t are independent Bernoulli random variables with $p_t = 1/n$, as in Theorem 9, and $\mu = \mathbb{E}[Y_i] = T/n = 24 \log n$.
The lower tail bound of Theorem 9 with $\delta = 1/2$, for which $(1-\delta)\mu = 12 \log n$, gives for every slot i

$$\mathbb{P}(Y_i < 12 \log n) \leqslant e^{-\frac{\mu\delta^2}{2}} = e^{-\frac{24 \log n}{8}} = \frac{1}{n^3}.$$

By the union bound over the n slots, some slot has $Y_i < 12 \log n$ with probability at most $n \cdot 1/n^3 = 1/n^2$. For $n \geqslant 2$ we have $\log n > 0$, so $Y_i \leqslant 6 \log n$ implies $Y_i < 12 \log n$. So the second event is contained in the first and also has probability at most $1/n^2$. □

Week: 2 (August 31 – September 4, 2026)

Lecture Date: Thursday, September 3, 2026

1 Markov's Inequality

Markov's inequality uses only the expectation $\mathbb{E}[X]$ of the random variable, which is also called the *first moment*. Figure 3 shows the bound on the number line.

Theorem 10 ▶ Markov's Inequality

Let $X \geqslant 0$ be a random variable with $\mathbb{E}[X] > 0$. Then, for any $a > 0$,

$$\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}.$$

Equivalently, for any $b > 0$,

$$\mathbb{P}(X \geqslant b) \leqslant \frac{\mathbb{E}[X]}{b}.$$

Proof. Let $I = 1$ if $X \geqslant b$ and $I = 0$ otherwise. Then $I \leqslant \frac{X}{b}$: if $X \geqslant b$ then $I = 1 \leqslant \frac{X}{b}$, and if $X < b$ then $I = 0 \leqslant \frac{X}{b}$ because $X \geqslant 0$. So

$$\mathbb{P}(X \geqslant b) = \mathbb{P}(I = 1) = \mathbb{E}[I] \leqslant \mathbb{E}\left[\frac{X}{b}\right] = \frac{\mathbb{E}[X]}{b}.$$

Taking $b = a\,\mathbb{E}[X] > 0$ gives $\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}$. □

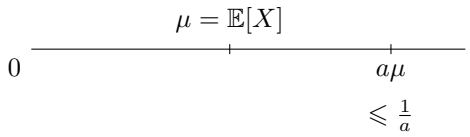


Figure 3: Markov's inequality: X is at least $a\mu$ with probability at most $1/a$.

2 Chebyshev's Inequality and the Chernoff Bound

Chebyshev's inequality uses the variance of the random variable, which is also called the *second moment*.

Definition 8 ▶ Variance and Standard Deviation

The **variance** of a random variable X is $\mathbb{V}[X] = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]$. The **standard deviation** of X is $\sigma = \sqrt{\mathbb{V}[X]}$.

Theorem 11 ▶ Chebyshev's Inequality

For any random variable X and any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) \leqslant \frac{\mathbb{V}[X]}{a^2} = \frac{\sigma^2}{a^2}.$$

Proof. Both sides of $|X - \mathbb{E}[X]| \geqslant a$ are nonnegative, so we can square them:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) = \mathbb{P}\Big((X - \mathbb{E}[X])^2 \geqslant a^2\Big) \leqslant \frac{\mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]}{a^2} = \frac{\mathbb{V}[X]}{a^2},$$

by Markov's inequality (Theorem 10) for $(X - \mathbb{E}[X])^2 \geqslant 0$. □

Chernoff bound. The Chernoff bounds (Theorem 9) need $X = \sum_{i=1}^{n} X_i$ to be a sum of *mutually independent* indicator random variables. Then $\mu = \mathbb{E}[X] = \sum_{i=1}^{n} p_i$ with $p_i = \mathbb{P}(X_i = 1)$ by linearity of expectation.

3 Search

Let $f\colon D \to R$ be a function from a domain D to a range R. Given y, we ask: is there an x such that $f(x) = y$ (Figure 4)? We consider $f\colon [1, \ldots, N] \to [1, \ldots, N]$ and want to find an x such that $f(x) = k$. The function is an array with n slots (Figure 1), where n is the size of the domain, and the cost of an algorithm is the number of queries to f. A randomized classical algorithm needs $\Theta(n)$ queries, and Grover's algorithm needs $\mathcal{O}(\sqrt{n})$ queries. For the random sampling algorithm, $\mathbb{E}[\text{queries}] = \mathcal{O}(n)$ and one sample has $\mathbb{P}(\text{Success}) = 1/n$.

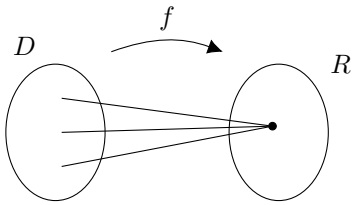


Figure 4: A function $f\colon D \to R$; several x can have the same value $y = f(x)$.

Probability amplitude amplification. If the classical success probability is p, then in the quantum case it becomes $\sqrt{p}$.

4 Birthday Paradox

There are m people and n birthdays ($n = 365$), and $f(x)$ is the birthday of person x. A *collision* is a pair $x \neq y$ with $f(x) = f(y)$.

Let E_i be the event that the ith birthday is different from the first $i - 1$ birthdays. By the chain rule and $1 - x \leqslant e^{-x}$,

$$\mathbb{P}(\text{all birthdays different}) = \mathbb{P}(E_1) \cdot \mathbb{P}(E_2 \mid E_1) \cdot \mathbb{P}(E_3 \mid E_2 \cap E_1) \cdots = \prod_{i=0}^{m-1} \left(1 - \frac{i}{n}\right)$$
$$\leqslant \prod_{i=0}^{m-1} e^{-i/n} = e^{-\frac{(m-1)m}{2n}}.$$

For $m = \sqrt{2n} + 1$ we have $(m-1)m \geqslant 2n$, so this is at most $1/e \leqslant 1/2$. So $\mathcal{O}(\sqrt{n})$ people are enough to have a constant probability of collision.

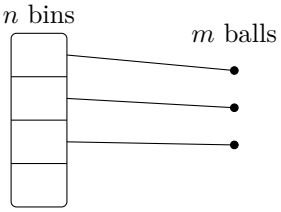


Figure 5: Throwing m balls into n bins.

4.1 Expected Number of Collisions

The birthday paradox can be seen as throwing m balls into n bins (Figure 5). For $m \geqslant \sqrt{2n}+1$, $\mathbb{E}[\#\text{ collisions}] \geqslant 1$.

Let X be the number of collisions. X takes values from 0 to $\binom{m}{2}$. Write

$$X = \sum_{i \neq j} X_{ij}, \qquad X_{ij} = \begin{cases} 1 & \text{if } f(i) = f(j), \\ 0 & \text{else,} \end{cases}$$

where the sum is over the $\binom{m}{2}$ pairs $i \neq j$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i \neq j} \mathbb{P}(X_{ij} = 1) = \binom{m}{2}\frac{1}{n} = \frac{m(m-1)}{2n} \sim \frac{m^2}{2n} = \Theta\left(\frac{m^2}{n}\right).$$

For $m \geqslant \sqrt{2n} + 1$ we have $m(m-1) \geqslant 2n$, so $\mathbb{E}[X] \geqslant 1$.

Claim 12 (Variance of the Number of Collisions)**.** *The X_{ij} are* pairwise independent*: for any two different pairs $\{i,j\} \neq \{k,l\}$,*

$$\mathbb{E}[X_{ij}X_{kl}] = \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}] = \frac{1}{n^2}.$$

So

$$\mathbb{V}[X] = \sum_{i \neq j} \mathbb{V}[X_{ij}] = \binom{m}{2}\frac{1}{n}\left(1 - \frac{1}{n}\right) \leqslant \mathbb{E}[X].$$

Proof. We have $\mathbb{E}[X_{ij}X_{kl}] = \mathbb{P}(f(i) = f(j) \text{ and } f(k) = f(l))$. If the pairs are disjoint, the two events depend on four independent birthdays, so the probability is $1/n^2$. If they share one person, say $\{i,j\}$ and $\{i,k\}$, then for every value b of $f(i)$ each of $f(j)$ and $f(k)$ equals b with probability $1/n$, independently, so the probability is again $1/n^2$. Two indicators with $\mathbb{E}[X_{ij}X_{kl}] = \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]$ are independent.

Expand the square of $X - \mathbb{E}[X] = \sum_{i \neq j}(X_{ij} - \mathbb{E}[X_{ij}])$:

$$\begin{aligned} \mathbb{V}[X] &= \sum_{i \neq j} \mathbb{E}\Big[(X_{ij} - \mathbb{E}[X_{ij}])^2\Big] + \sum_{\{i,j\} \neq \{k,l\}} \mathbb{E}[(X_{ij} - \mathbb{E}[X_{ij}])(X_{kl} - \mathbb{E}[X_{kl}])] \\ &= \sum_{i \neq j} \mathbb{V}[X_{ij}] + \sum_{\{i,j\} \neq \{k,l\}} (\mathbb{E}[X_{ij}X_{kl}] - \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]) = \sum_{i \neq j} \mathbb{V}[X_{ij}], \end{aligned}$$

because the second sum is zero by the first part. Since $X_{ij}^2 = X_{ij}$, $\mathbb{V}[X_{ij}] = \mathbb{E}\Big[X_{ij}^2\Big] - \mathbb{E}[X_{ij}]^2 = \frac{1}{n}\Big(1 - \frac{1}{n}\Big) \leqslant \frac{1}{n}$. Summing over the $\binom{m}{2}$ pairs gives $\mathbb{V}[X] \leqslant \binom{m}{2}\frac{1}{n} = \mathbb{E}[X]$. □

Exercise 2. Show that the birthday paradox occurs: with $\mathcal{O}(\sqrt{n})$ people, $\mathbb{P}(X \geqslant 1) \geqslant \Theta(1)$.

Precisely, if $m \geqslant 2\sqrt{n}+1$, then

$$\mathbb{P}(X \geqslant 1) \geqslant \frac{1}{2}.$$

Proof. If $X = 0$ then $|X - \mathbb{E}[X]| = \mathbb{E}[X]$. So by Chebyshev's inequality (Theorem 11) with $a = \mathbb{E}[X]$, and then Claim 12,

$$\mathbb{P}(X = 0) \leqslant \mathbb{P}(|X - \mathbb{E}[X]| \geqslant \mathbb{E}[X]) \leqslant \frac{\mathbb{V}[X]}{\mathbb{E}[X]^2} \leqslant \frac{1}{\mathbb{E}[X]}.$$

For $m \geqslant 2\sqrt{n}+1$, since $m(m-1)$ grows with m, $\mathbb{E}[X] = \frac{m(m-1)}{2n} \geqslant \frac{(2\sqrt{n}+1)\cdot 2\sqrt{n}}{2n} \geqslant 2$, so $\mathbb{P}(X = 0) \leqslant 1/2$ and $\mathbb{P}(X \geqslant 1) = 1 - \mathbb{P}(X = 0) \geqslant 1/2$. □

5 Quantum Birthday Paradox

The *collision problem* is: given query access to a function f, find $x \neq y$ with $f(x) = f(y)$. By Exercise 2, $\mathcal{O}(\sqrt{n})$ classical queries to random points are enough. An algorithm makes k queries $f(x_1), \ldots, f(x_k)$, where x_s may depend on the earlier answers and on random coins.

As in the birthday paradox, $f\colon D \to [n]$ is a *random function*: every value $f(x)$ is uniform in $[n] = \{1, \ldots, n\}$ and independent of the others. We assume that the algorithm queries the two points that it outputs (at most 2 extra queries) and never repeats a query.

Exercise 3. Show a $\sqrt{n}$ lower bound in the classical case: for a random function $f\colon D \to [n]$, every classical algorithm (deterministic or randomized, adaptive) that makes k queries to different points, including the two points that it outputs, finds a collision with probability at most

$$\frac{\binom{k}{2}}{n} = \frac{k(k-1)}{2n}.$$

So to find a collision with probability at least $1/2$ it needs $k > \sqrt{n}$ queries, and the classical query complexity of the collision problem is $\Omega(\sqrt{n})$.

Proof. A collision needs two queried values to agree. Let E_s be the event that $f(x_s) \in \{f(x_1), \ldots, f(x_{s-1})\}$. Given the first $s-1$ answers and the random coins, x_s is a fixed point that has not been queried, so $f(x_s)$ is uniform in $[n]$ and independent of the $s-1$ earlier values. So $\mathbb{P}(E_s) \leqslant \frac{s-1}{n}$ for every such outcome, and so also without conditioning. By the union bound,

$$\mathbb{P}(\text{collision}) \leqslant \sum_{s=1}^{k} \mathbb{P}(E_s) \leqslant \sum_{s=1}^{k} \frac{s-1}{n} = \frac{k(k-1)}{2n}.$$

If the algorithm succeeds with probability at least $1/2$, then $\frac{k(k-1)}{2n} \geqslant \frac{1}{2}$, so $k^2 > k(k-1) \geqslant n$ and $k > \sqrt{n}$. □

Week: 3 (September 7–11, 2026)

Lecture Date: Tuesday, September 8, 2026

1 Distributed Computing in a Complete Network

The network is a *complete graph* G on n nodes (Figure 6). Every pair of nodes is joined by a bidirectional communication link, so there are $\binom{n}{2} = \frac{n(n-1)}{2}$ edges and every node has $n-1$ incident edges. The network is *synchronous*: all nodes share a clock, and the computation runs in *rounds*. In one round, every node receives the messages sent to it in the previous round, does a local computation, and sends messages on some of its edges (receives → computes → sends).

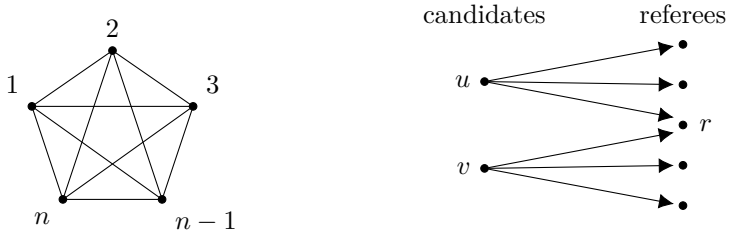


Figure 6: Left: the complete network on n nodes. Right: candidates u and v with a common referee r.

1.1 Leader Election

Problem 1 ▶ Leader Election

All nodes run the same algorithm. At the end, exactly one node must be in the state *leader*, and all other nodes must be in the state *non-leader*.

The goals are to minimize the number of rounds and the total number of messages sent. The nodes are either *labelled* (every node has a unique id) or *anonymous* (no ids).

The trivial algorithm. With unique ids, every node sends its id to all other nodes in the first round, and the node with the minimum id becomes the leader. This takes $\mathcal{O}(1)$ rounds and $n(n-1) = \Theta(n^2)$ messages.

Anonymous nodes. A node addresses its $n-1$ edges by *ports* $1, \dots, n-1$. It sees on which port a message arrives, but not which node is at the other end, and an adversary chooses the port numbering.

Exercise 4. Let $n \geqslant 2$. Prove that in an anonymous complete network in which all nodes start in the same state, there is no deterministic algorithm that elects a leader for every port numbering: for every deterministic algorithm there is a port numbering for which, after every number of rounds, the number of leaders is either 0 or n.

Proof. Fix a deterministic algorithm. Number the nodes by $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ (addition modulo n), and let port j of node i lead to node $i+j$, for $j = 1, \dots, n-1$. Then the edge between i and $i+j$ is port j at i and port $n-j$ at $i+j$, so a message sent on port j arrives

on port $n - j$.
By induction on t, at the end of round t all nodes are in the same state, and in round t all nodes send the same message m_j on port j ("no message" counts as a message). This holds for $t = 0$. In round t node i receives on port q the message that node $i + q$ sent on its port $n - q$, which is the common message m_{n-q} of round $t - 1$ for every i. So all nodes receive the same messages on the same ports. The algorithm is deterministic, so they move to the same state and send the same messages.
Whether a node is a leader depends only on its state. So after every round either all $n \geqslant 2$ nodes are leaders or none is. □

From now on we assume that the nodes have unique ids.

Complexity classes. Deterministic, bounded-error probabilistic and bounded-error quantum polynomial time satisfy $\mathrm{P} \subseteq \mathrm{BPP} \subseteq \mathrm{BQP}$.

1.2 Known Bounds

For leader election in a complete network of n nodes with unique ids, the following bounds are known ($\widetilde{O}$ and $\widetilde{\Omega}$ hide factors polynomial in $\log n$). We prove only the first and the third row.

Messages	Rounds	Algorithm
$\Theta(n^2)$	$\mathcal{O}(1)$	trivial, deterministic
$\mathcal{O}(n \log n)$	$\mathcal{O}(\log n)$	deterministic
$\mathcal{O}\left(\sqrt{n} \log^{3/2} n\right) = \widetilde{O}(\sqrt{n})$	$\mathcal{O}(1)$	randomized (this lecture)
$\widetilde{\Omega}(\sqrt{n})$		classical lower bound
$\widetilde{O}(n^{1/3})$		quantum

2 A Randomized Algorithm for Leader Election

In this section, log is the natural logarithm, $p = \frac{2 \log n}{n}$ and $k = \lceil \sqrt{n \log n} \rceil$. We have $p \leqslant 1$ and $k \leqslant n$ for every $n \geqslant 1$, since $2 \log n \leqslant n$.

1. **Subsample candidates.** Every node becomes a *candidate* with probability p, independently of all other nodes.
2. **Candidates contact referees.** Every candidate sends its id to k random nodes, its *referees*.
3. **Referee round.** Every node that received at least one id declares the minimum id among the received messages the *winner*, and replies with the winner's id to all candidates that contacted it.
4. **Decision.** A candidate becomes the leader if it is the winner at all of its referees. All other nodes are non-leaders.

The model for the random choices. Every node u draws a coin $C_u \in \{0, 1\}$ with $\mathbb{P}(C_u = 1) = p$ (u is a candidate if $C_u = 1$) and a *referee set* R_u, which is a uniformly random set of k distinct nodes among all n nodes. All the C_u and all the R_u are mutually independent. Only candidates use their referee sets. A candidate may pick itself as a referee. It then plays the role of this referee itself, and no message is needed.

2.1 The Number of Candidates

Let $X = \sum_u C_u$ be the number of candidates. By linearity of expectation, $\mathbb{E}[X] = np = 2\log n$.

Exercise 5. Show that there is at least one leader if and only if there is at least one candidate, and that

$$\mathbb{P}(\text{no leader}) = \mathbb{P}(X = 0) \leqslant \frac{1}{n^2}, \qquad \text{so} \qquad \mathbb{P}(\text{at least 1 leader}) \geqslant 1 - \frac{1}{n^2}.$$

Proof. Only candidates become leaders, so if $X = 0$ there is no leader. If $X \geqslant 1$, let w be the candidate with the minimum id (unique, since the ids are unique). Every referee r of w receives only ids of candidates. All of them are at least the id of w, and the id of w is one of them. So r declares w the winner, and w is a leader. The coins are independent, so, using $0 \leqslant 1 - p$ and $1 - x \leqslant e^{-x}$, $\mathbb{P}(X = 0) = (1-p)^n \leqslant e^{-pn} = e^{-2\log n} = 1/n^2$. □

2.2 At Most One Leader

A pair of different nodes $\{u, v\}$ is a *bad pair* if both are candidates and they have no common referee: $C_u = C_v = 1$ and $R_u \cap R_v = \emptyset$.

Claim 13 (Two Leaders Form a Bad Pair)**.** *If two candidates have a common referee, then at most one of them is a leader. So if there is more than one leader, then there is a bad pair.*

Proof. Let $u \neq v$ be candidates with a common referee r, where the id of u is smaller than the id of v (Figure 6). The referee r receives both ids (if r is u or v itself, it knows its own id), so the minimum id that it receives is at most the id of u. Then r does not declare v the winner, and v is not a leader. So two different leaders are two candidates without a common referee, which is a bad pair. □

Claim 14 (Two Referee Sets Rarely Miss Each Other)**.** *For any two different nodes u and v,*

$$\mathbb{P}(R_u \cap R_v = \emptyset) \leqslant \left(1 - \frac{k}{n}\right)^k \leqslant e^{-k^2/n} \leqslant \frac{1}{n}.$$

Proof. Condition on $R_u = S$, where S is any set of k nodes. By independence, R_v is still a uniformly random set of k nodes. If $2k > n$, two sets of k nodes cannot be disjoint and the probability is 0. So let $2k \leqslant n$. Draw the k nodes of R_v one after the other, each uniformly among the nodes not drawn yet. If the first i nodes drawn are outside S, then $n - i$ nodes are left and k of them are in S, so the next node is outside S with probability $\frac{n-k-i}{n-i}$. So

$$\mathbb{P}(R_v \cap S = \emptyset) = \prod_{i=0}^{k-1}\left(1 - \frac{k}{n-i}\right) \leqslant \left(1 - \frac{k}{n}\right)^k \leqslant e^{-k^2/n} \leqslant e^{-\log n} = \frac{1}{n},$$

because every factor satisfies $0 \leqslant 1 - \frac{k}{n-i} \leqslant 1 - \frac{k}{n}$ (since $i \leqslant k-1$ and $2k \leqslant n$ give $n - i \geqslant k$), and by $1 - x \leqslant e^{-x}$ and $k^2 \geqslant n\log n$. The bound does not depend on S, so it also holds without conditioning. □

Exercise 6. Show that

$$\mathbb{P}(\text{more than 1 leader}) \leqslant \frac{2\log^2 n}{n},$$

so there is at most 1 leader with probability tending to 1.

Proof. Let B be the number of bad pairs. For a fixed pair $\{u, v\}$, the coins C_u, C_v and the pair of sets (R_u, R_v) are independent, so by Claim 14 the pair is bad with probability $p^2 \cdot \mathbb{P}(R_u \cap R_v = \emptyset) \leqslant p^2/n$. By linearity of expectation over the $\binom{n}{2}$ pairs, $\mathbb{E}[B] \leqslant \binom{n}{2}\frac{p^2}{n} = \frac{n(n-1)}{2} \cdot \frac{4\log^2 n}{n^3} \leqslant \frac{2\log^2 n}{n}$. If there is more than one leader, then $B \geqslant 1$ by Claim 13. Since $B \geqslant 0$, Markov's inequality (Theorem 10) with $b = 1$ gives $\mathbb{P}(\text{more than 1 leader}) \leqslant \mathbb{P}(B \geqslant 1) \leqslant \mathbb{E}[B]$, and $2\log^2 n/n \to 0$ as $n \to \infty$. □

2.3 Messages and Rounds

Theorem 15 ▶ Complexity of the Algorithm

The algorithm takes 3 rounds, and messages are sent only in the first two. The expected number of messages is at most $4k \log n = \mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$.

Proof. Rounds. In round 1 every node draws its coin and its referee set, and the candidates send their ids. In round 2 the referees receive the ids and reply with the minimum. In round 3 the candidates receive the replies and decide, and no message is sent.

Messages. Every candidate sends at most k messages and gets at most k replies, so the number of messages is at most $2kX$. Its expectation is at most $2k\,\mathbb{E}[X] = 4k \log n$, which is $\mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$ since $k \leqslant \sqrt{n \log n} + 1$. □

3 Markov Chains: Random Walks in Undirected Graphs

Definition 9 ▶ Random Walk

Let G be an undirected graph in which every node has at least one neighbour. In a **random walk** on G, a walker starts at some node. In every step it moves from its current node u to each neighbour of u with probability $\frac{1}{\deg(u)}$, independently of the earlier steps.

Let $p_t = (p_t^1, \ldots, p_t^n)$ be the row vector where p_t^i is the probability that the walker is at node i after t steps. The walker is at one node, so the entries of p_t sum to 1. If the walker is at node j after t steps, it moves to a given neighbour i of j with probability $\frac{1}{\deg(j)}$. So by the law of total probability,

$$p_{t+1}^i = \sum_{j\,:\,j \text{ is a neighbour of } i} \frac{p_t^j}{\deg(j)}. \tag{1}$$

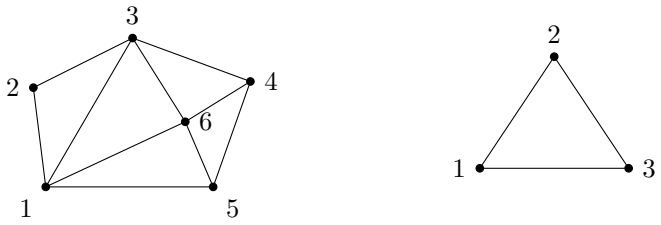


Figure 7: Left: a graph with 6 nodes and 10 edges. Right: the triangle.

Example 1. In the graph on the left of Figure 7, a walker that starts at node 1 has $p_0 = (1, 0, 0, 0, 0, 0)$ and, since node 1 has the 4 neighbours 2, 3, 5, 6, $p_1 = \left(0, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}\right)$. In

the triangle on the right every node has degree 2, and starting at node 1, by Equation (1), $p_0 = (1, 0, 0)$, $p_1 = \left(0, \frac{1}{2}, \frac{1}{2}\right)$ and $p_2 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$.

4 The Law of Large Numbers

Exercise 7. Let $X_1, X_2, \ldots$ be pairwise independent random variables with the same expectation $\mathbb{E}[X_i] = \mu$ and the same finite variance $\mathbb{V}[X_i] = \sigma^2$, and let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Show the weak law of large numbers with Chebyshev's inequality: for every $\varepsilon > 0$,

$$\mathbb{P}\Big(\Big|\overline{X}_n - \mu\Big| \geqslant \varepsilon\Big) \leqslant \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Proof. By linearity of expectation, $\mathbb{E}\Big[\overline{X}_n\Big] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \mu$. Let $Y_i = X_i - \mu$. Then $\mathbb{E}[Y_i] = 0$, $\mathbb{E}[Y_i^2] = \sigma^2$, and $\overline{X}_n - \mu = \frac{1}{n}\sum_i Y_i$. For $i \neq j$, Y_i and Y_j are independent, so $\mathbb{E}[Y_i Y_j] = \mathbb{E}[Y_i]\,\mathbb{E}[Y_j] = 0$. Expanding the square as in the proof of Claim 12 gives

$$\mathbb{V}\Big[\overline{X}_n\Big] = \mathbb{E}\left[\Big(\frac{1}{n}\sum_{i=1}^{n} Y_i\Big)^2\right] = \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\Big[Y_i^2\Big] + \frac{1}{n^2}\sum_{i\neq j}\mathbb{E}[Y_i Y_j] = \frac{\sigma^2}{n}.$$

Chebyshev's inequality (Theorem 11) for $\overline{X}_n$ with $a = \varepsilon$ gives $\mathbb{P}\Big(\Big|\overline{X}_n - \mu\Big| \geqslant \varepsilon\Big) \leqslant \mathbb{V}\Big[\overline{X}_n\Big]/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$, which tends to 0 as $n \to \infty$. □

Week: 3 (September 7–11, 2026)

Lecture Date: Thursday, September 10, 2026

Law of large numbers. We proved the weak form in Exercise 7.

Theorem 16 ▶ Law of Large Numbers

Let $X_1, X_2, \dots$ be independent, identically distributed (i.i.d.) random variables distributed as X, where $\mathbb{E}[X]$ is finite. Then, with probability 1,

$$\lim_{n\to\infty} \frac{\sum_{i=1}^{n} X_i}{n} = \mathbb{E}[X].$$

1 Random Walks on Graphs

Let $G = (V, E)$ be a finite undirected graph without isolated vertices, with $V = \{1, \dots, n\}$ and $m = |E|$ edges. We write $u \sim v$ if $\{u, v\} \in E$, and $\deg(u)$ (or d_u) for the number of neighbours of u. The walker is now called the token, and X_t is its vertex at time $t = 0, 1, 2, \dots$.

Definition 10 ▶ Distribution at Time t, Transition Matrix

The **distribution at time** t is the row vector

$$p_t = \left(p_t^1, p_t^2, \dots, p_t^n\right), \qquad p_t^i = \mathbb{P}(X_t = i).$$

The **transition matrix** is the $n \times n$ matrix P with entries

$$P_{uv} = \mathbb{P}(u \to v) = \mathbb{P}(X_{t+1} = v \mid X_t = u) = \begin{cases} \frac{1}{\deg(u)} & \text{if } u \sim v, \\ 0 & \text{otherwise.} \end{cases}$$

1.1 The Triangle

In the triangle on the vertices $1, 2, 3$ (the right half of Figure 7) every vertex has degree 2, so P is the 3×3 matrix with 0 on the diagonal and $1/2$ in every other entry.

1.2 The Evolution $p_{t+1} = p_t P$

Lemma 17 ▶ Evolution of the Distribution

For every $t \geqslant 0$,

$$p_{t+1} = p_t P, \qquad \text{and so} \qquad p_t = p_0 P^t.$$

Proof. Fix a vertex v. The events $\{X_t = u\}$, $u \in V$, are disjoint and one of them occurs, so

$$p_{t+1}^v = \mathbb{P}(X_{t+1} = v) = \sum_{u \in V} \mathbb{P}(X_t = u \text{ and } X_{t+1} = v) = \sum_{u \in V} p_t^u P_{uv} = (p_t P)_v.$$

Here $\mathbb{P}(X_t = u \text{ and } X_{t+1} = v) = p_t^u P_{uv}$ by the definition of P_{uv} as a conditional probability when $p_t^u > 0$, and both sides are 0 when $p_t^u = 0$. So $p_{t+1} = p_t P$. Induction on t (with $P^0 = I$) gives $p_t = p_0 P^t$. □

Entry by entry, $p_{t+1} = p_t P$ is Equation (1). For the triangle with $p_0 = (1, 0, 0)$, p_1 and p_2 are in Example 1.

1.3 Row Stochastic Matrices

Definition 11 ▶ Probability Vector, Row Stochastic Matrix

A vector $p = (p^1, \ldots, p^n)$ is a **probability vector** if $p^u \geqslant 0$ for all u and $\sum_{u=1}^n p^u = 1$. An $n \times n$ matrix P is **row stochastic** if every row of P is a probability vector:

$$P_{uv} \geqslant 0 \text{ for all } u, v, \qquad \text{and} \qquad \sum_{v=1}^{n} P_{uv} = 1 \text{ for all } u.$$

The transition matrix of a random walk on any graph is row stochastic: row u has $\deg(u)$ entries equal to $1/\deg(u)$, and its other entries are 0.

Exercise 8. Let P be a row stochastic $n \times n$ matrix and let p be a probability vector. Show that pP is a probability vector. So if p_0 is a probability vector and $p_{t+1} = p_t P$ for all $t \geqslant 0$, then $p_t = p_0 P^t$ is a probability vector for every $t \geqslant 0$.

Proof. Let $q = pP$, so $q^v = \sum_{u=1}^n p^u P_{uv}$ for every v. Every term is a product of two nonnegative numbers, so $q^v \geqslant 0$. Every row of P and the entries of p sum to 1, so $\sum_v q^v = \sum_v \sum_u p^u P_{uv} = \sum_u p^u (\sum_v P_{uv}) = \sum_u p^u = 1$. So pP is a probability vector. The vector p_0 is one by assumption, and if p_t is one, then so is $p_t P = p_{t+1}$. By induction on t, every p_t is a probability vector. □

1.4 The Transition Matrix of a General Graph

Let A be the *adjacency matrix* of G, with $A_{uv} = 1$ if $u \sim v$ and $A_{uv} = 0$ otherwise. Let D be the diagonal matrix with diagonal entries $d_1, \ldots, d_n$. Its inverse D^{-1} has diagonal entries $1/d_1, \ldots, 1/d_n$ and exists because there are no isolated vertices. Then $(D^{-1}A)_{uv} = A_{uv}/\deg(u)$, which is P_{uv} of Definition 10, so $P = D^{-1}A$. If G is *d-regular*, meaning every vertex has the same degree d, then $D = dI$ and $P = A/d$. The triangle is 2-regular, and its P is $A/2$.

2 Stationary Distribution

Definition 12 ▶ Stationary Distribution

A probability vector π is a **stationary distribution** of the walk with transition matrix P if

$$\pi P = \pi \cdot 1 = \pi.$$

Then π is a left eigenvector of P with eigenvalue 1.

By Lemma 17, if $p_t = \pi$ then $p_{t+1} = \pi P = \pi$. Only the neighbours v of u have $P_{vu} \neq 0$, and for them $P_{vu} = 1/\deg(v)$, so (as in Equation (1))

$$(\pi P)_u = \sum_{v \in V} \pi_v P_{vu} = \sum_{v \sim u} \frac{\pi_v}{\deg(v)}. \tag{2}$$

The sum runs over the neighbours of u, as in Figure 8.

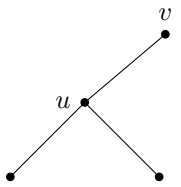


Figure 8: A vertex u and its neighbours; v is one of them.

Theorem 18 ▶ Stationary Distribution of a Random Walk on a Graph

Let G be a graph with m edges and no isolated vertices. Then

$$\pi_v = \frac{\deg(v)}{2m} \qquad (v \in V)$$

is a stationary distribution of the random walk on G.

Proof. π is a probability vector. There are no isolated vertices and V is not empty, so $m \geqslant 1$. So π is well defined, and $\pi_v \geqslant 0$. Every edge $\{u, v\}$ is counted once in $\deg(u)$ and once in $\deg(v)$. So $\sum_{v \in V} \deg(v) = 2m$ (the handshake lemma) and $\sum_{v \in V} \pi_v = 1$.

$\pi P = \pi$. For every vertex u, by Equation (2) and $\deg(v) \geqslant 1$, $(\pi P)_u = \sum_{v \sim u} \frac{\deg(v)}{2m} \cdot \frac{1}{\deg(v)} = \sum_{v \sim u} \frac{1}{2m} = \frac{\deg(u)}{2m} = \pi_u$, since u has $\deg(u)$ neighbours. □

3 Convergence to the Stationary Distribution

Convergence of vectors is entry by entry. We state the theorem without proof.

Theorem 19 ▶ Convergence of Random Walks

Let G be a connected, non-bipartite graph with m edges. Then for every starting distribution p_0,

$$p_t = p_0 P^t \longrightarrow \pi \qquad \text{as } t \to \infty, \qquad \text{where } \pi_v = \frac{\deg(v)}{2m}.$$

So π is the only stationary distribution of the walk.

Week: 4 (September 14–18, 2026)

Lecture Date: Tuesday, September 15, 2026

1 Qubits and Dirac Notation

A classical *bit* takes one of the two values 0 and 1. A quantum bit can be in a combination of both.

Definition 13 ▶ Qubit, n-Qubit State

The state of one **qubit** is a unit vector in $\mathbb{C}^2$,

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \alpha\begin{pmatrix}1\\ 0\end{pmatrix} + \beta\begin{pmatrix}0\\ 1\end{pmatrix} = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

where $|0\rangle = \begin{pmatrix}1\\ 0\end{pmatrix}$ and $|1\rangle = \begin{pmatrix}0\\ 1\end{pmatrix}$ form the **computational basis**. The complex numbers α, β are the **amplitudes**. More generally, the state of n qubits is a unit vector in $\mathbb{C}^{2^n}$,

$$|\phi\rangle = \sum_{x\in\{0,1\}^n} \alpha_x|x\rangle, \qquad \sum_{x\in\{0,1\}^n} |\alpha_x|^2 = 1,$$

where $|x\rangle$ is the standard basis vector with a 1 in the position indexed by the bit string x (the 2^n strings are listed in increasing binary order) and 0 elsewhere. Measuring the state gives the outcome x with probability $|\alpha_x|^2$.

Two qubits. The state of two qubits has four amplitudes: $(\alpha, \beta, \gamma, \delta)^T = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$ with $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$, where $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are the four standard basis column vectors of $\mathbb{C}^4$, in this order.

Exercise 9. What is the third entry of the column vector $|00\rangle$?

Solution. It is 0. The third entry is the amplitude of $|10\rangle$, and $|00\rangle = 1\cdot|00\rangle + 0\cdot|01\rangle + 0\cdot|10\rangle + 0\cdot|11\rangle$, so $|00\rangle = (1,0,0,0)^T$. □

Writing a column vector as a combination of the kets $|x\rangle$ is *Dirac notation*. The state $\frac{1}{\sqrt{2}}|0000\rangle + \frac{1}{\sqrt{2}}|1111\rangle$ of four qubits is a column vector with $2^4 = 16$ entries: $1/\sqrt{2}$ in the first and in the last position and 0 in the other fourteen.

1.1 Kets, Bras and the Dagger

A *ket* $|\phi\rangle$ is a column vector. The corresponding *bra* is the row vector $\langle\phi| = (|\phi\rangle)^\dagger$, where † ("dagger") is the *conjugate transpose*: transpose and replace every entry by its complex conjugate. For $|\phi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ we get $\langle\phi| = \begin{pmatrix}\overline{\alpha} & \overline{\beta}\end{pmatrix}$. The dagger equals the inverse only for unitary matrices (Definition 14).

A bra times a ket is a number, the *inner product*, and it gives the (Euclidean) norm. For

$|x\rangle, |y\rangle \in \mathbb{C}^n$ with entries x_i, y_i,

$$\langle x|y\rangle = \langle x||y\rangle = \sum_{i=1}^{n} \overline{x_i}\, y_i, \qquad \||y\rangle\|^2 = \langle y|y\rangle = \sum_{i=1}^{n} |y_i|^2.$$

The inner product is linear in the second argument and conjugate-linear in the first, and $\langle y|x\rangle = \overline{\langle x|y\rangle}$. We use two rules for the dagger: $(A^\dagger)^\dagger = A$, and $(AB)^\dagger = B^\dagger A^\dagger$ for matrices A, B whose product is defined. So $(U|x\rangle)^\dagger = \langle x|U^\dagger$.

2 Unitary Matrices

Definition 14 ▶ Unitary Matrix

An $n \times n$ complex matrix U is **unitary** if $U^\dagger U = I$.

Exercise 10. Let U be an $n \times n$ complex matrix. Show that $\|U|x\rangle\| = \||x\rangle\|$ for all $|x\rangle \in \mathbb{C}^n$ if and only if U is unitary ($U^\dagger U = I$).

Proof. Unitary implies norm preserving. If $U^\dagger U = I$, then for every $|x\rangle$, $\|U|x\rangle\|^2 = (U|x\rangle)^\dagger U|x\rangle = \langle x|U^\dagger U|x\rangle = \langle x|x\rangle = \||x\rangle\|^2$. Norms are non-negative, so $\|U|x\rangle\| = \||x\rangle\|$.
Norm preserving implies unitary. Let $M = U^\dagger U - I$. Then $M^\dagger = M$, and for every $|x\rangle$,

$$\langle x|M|x\rangle = \|U|x\rangle\|^2 - \||x\rangle\|^2 = 0.$$

Let $z = \langle x|M|y\rangle$, so that $\langle y|M|x\rangle = \overline{z}$. Applying the last equation to $|x\rangle + |y\rangle$ and to $|x\rangle + i|y\rangle$ gives

$$0 = z + \overline{z} = 2\,\mathrm{Re}\, z, \qquad 0 = i(z - \overline{z}) = -2\,\mathrm{Im}\, z.$$

So $\langle x|M|y\rangle = 0$ for all $|x\rangle, |y\rangle$. For the basis vectors e_j and e_k this says $M_{jk} = 0$. So $M = 0$, which is $U^\dagger U = I$. □

Remark. A quantum gate is a linear map that sends every state (every unit vector) to a state. Every vector is a multiple of a unit vector, so such a map preserves the norm of all vectors. By Exercise 10 it is unitary. Conversely, every unitary matrix maps states to states.

3 Quantum Circuits and One-Qubit Gates

A *quantum circuit* is drawn with one horizontal wire per qubit. Time runs from left to right, and a box on a wire is a gate (a unitary matrix) applied to that qubit (Figure 9).

Columns of a matrix. Let A be an $m \times n$ matrix with columns $a_1, \ldots, a_n$, and let $x = (x_1, \ldots, x_n)^T$. Then $Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n$. So $Ae_j = a_j$: *the jth column of a gate's matrix is the image of the jth basis state.*

The X gate. The *Pauli X gate* is the quantum NOT gate, with $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. By linearity,

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad X(\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle + \beta|0\rangle.$$

Definition 15 ▶ Hadamard Gate

The **Hadamard gate** H is the one-qubit gate with

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{3}$$

Both $H|0\rangle$ and $H|1\rangle$ are equal superpositions: measuring gives 0 and 1 with probability $1/2$ each.

Lemma 20 ▶ Hadamard Twice Is the Identity

$H^2 = I$. So H is unitary, and

$$H\,\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle, \qquad H\,\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |1\rangle.$$

Proof. The diagonal entries of H^2 are $\frac{1}{2}(1\cdot 1 + 1\cdot 1) = 1$ and $\frac{1}{2}(1\cdot 1 + (-1)(-1)) = 1$, and both off-diagonal entries are $\frac{1}{2}(1\cdot 1 + 1\cdot(-1)) = 0$, so $H^2 = I$. H is real and symmetric, so $H^\dagger = H$ and $H^\dagger H = H^2 = I$. By Equation (3), the two states in the statement are $H|0\rangle$ and $H|1\rangle$, and $HH|0\rangle = |0\rangle$, $HH|1\rangle = |1\rangle$. □

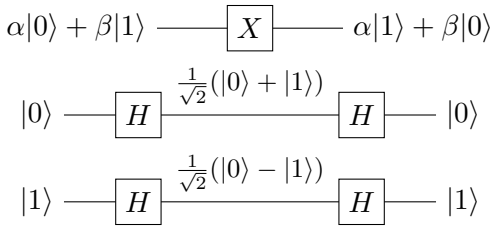


Figure 9: Top: the X gate. Below: the Hadamard gate applied twice (Lemma 20).

4 The CNOT Gate

Definition 16 ▶ CNOT Gate

The **controlled-NOT gate** (CNOT) is the two-qubit gate that maps the basis states by

$$|a\rangle|b\rangle \longmapsto |a\rangle|b \oplus a\rangle, \qquad a, b \in \{0, 1\},$$

where $\oplus$ is XOR (addition modulo 2). The first qubit $|a\rangle$ is the **control** qubit and is left unchanged. The second qubit $|b\rangle$ is the **target** qubit and is flipped exactly when $a = 1$. See Figure 10.

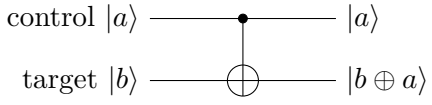


Figure 10: The CNOT gate: a filled dot on the control qubit and $\oplus$ on the target qubit.

Lemma 21 ▶ Matrix of the CNOT Gate

The matrix of the CNOT gate, with rows and columns indexed by $00, 01, 10, 11$, is the permutation matrix

$$U_{\text{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

It is unitary.

Proof. The columns of the matrix of a gate are the images of the basis states. By Definition 16, $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ go to $|00\rangle, |01\rangle, |11\rangle, |10\rangle$, which are the four columns of U_{CNOT} in this order. They are the four standard basis vectors, each once, so U_{CNOT} is a *permutation matrix*.

A permutation matrix P is unitary: the (j, k) entry of $P^T P$ is the inner product of the jth and the kth column of P, which is 1 if $j = k$ and 0 otherwise. So $P^\dagger P = P^T P = I$. □

In block form, $U_{\text{CNOT}} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}$. If the control is 0 nothing happens, and if it is 1 the gate X is applied to the target.

4.1 Classical Gates Are Not Reversible

By Exercise 10 and the remark after it, every quantum gate is unitary, so applying $U^\dagger$ undoes it. Classical gates that take two bits to one bit, such as AND, OR and XOR, are *not bijective*, so *not reversible*, and so *not unitary*.

Lemma 22 ▶ XOR Is Not Reversible

The classical gate XOR: $(a, b) \mapsto a \oplus b$ is not injective, so its input cannot be recovered from its output. The linear map $T\colon \mathbb{C}^4 \to \mathbb{C}^2$ with $T|ab\rangle = |a \oplus b\rangle$ does not preserve norms, so it is not unitary. The same holds for AND and OR.

Proof. $\text{XOR}(0, 0) = 0 = \text{XOR}(1, 1)$, and $T(|00\rangle - |11\rangle) = |0\rangle - |0\rangle = 0$ although $\||00\rangle - |11\rangle\| = \sqrt{2}$. For AND use the inputs 00 and 01, and for OR use 01 and 11. □

5 Tensor Products and Entanglement

Definition 17 ▶ Tensor Product

Let $A = (a_{ij})$ be an $m \times n$ matrix and let B be a $p \times q$ matrix. The **tensor product** (Kronecker product) $A \otimes B$ is the $mp \times nq$ block matrix

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & & & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}.$$

Column vectors are matrices with one column, so

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}.$$

For example, $|1\rangle \otimes |0\rangle = (0,0,1,0)^T = |10\rangle$, and in general $|a\rangle|b\rangle = |a\rangle \otimes |b\rangle = |ab\rangle$ for $a, b \in \{0,1\}$. In terms of kets, $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$. If A is applied to the first qubit and B to the second, the two-qubit gate is $A \otimes B$, and

$$(A \otimes B)(|x\rangle \otimes |y\rangle) = A|x\rangle \otimes B|y\rangle. \tag{4}$$

If $x_1, \ldots, x_n$ are the entries of $|x\rangle$, then $|x\rangle \otimes |y\rangle$ consists of the blocks $x_1|y\rangle, \ldots, x_n|y\rangle$. So the ith block of the left-hand side is $\sum_j a_{ij} B(x_j|y\rangle) = \left(\sum_j a_{ij}x_j\right) B|y\rangle$, which is the ith block of the right-hand side.

Definition 18 ▶ Product State, Entangled State

A two-qubit state $|\psi\rangle$ is a **product state** if $|\psi\rangle = |\phi_1\rangle \otimes |\phi_2\rangle$ for some one-qubit states $|\phi_1\rangle$, $|\phi_2\rangle$. Otherwise $|\psi\rangle$ is **entangled**.

Exercise 11. Show that the *Bell state* $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ is entangled: there are no vectors $|\phi_1\rangle, |\phi_2\rangle \in \mathbb{C}^2$ with

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = |\phi_1\rangle \otimes |\phi_2\rangle.$$

Proof. Suppose there are, and write $|\phi_1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $|\phi_2\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$. By Definition 17, comparing the four entries of the two sides gives $\alpha\gamma = 1/\sqrt{2}$, $\alpha\delta = 0$, $\beta\gamma = 0$ and $\beta\delta = 1/\sqrt{2}$. From $\alpha\delta = 0$ we get $\alpha = 0$ or $\delta = 0$. If $\alpha = 0$ then $\alpha\gamma = 0 \neq 1/\sqrt{2}$, and if $\delta = 0$ then $\beta\delta = 0 \neq 1/\sqrt{2}$. Both cases are contradictions. □

Week: 4 (September 14–18, 2026)

Lecture Date: Thursday, September 17, 2026

1 The Product Ax as a Combination of Columns

Let $A \in \mathbb{R}^{3\times 3}$ be the matrix with columns $v_1 = (2,3,5)^T$, $v_2 = (1,1,7)^T$, $v_3 = (3,4,12)^T$. Then

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 4 \\ 5 & 7 & 12 \end{bmatrix}, \qquad Ax = \begin{bmatrix} 2x_1 + x_2 + 3x_3 \\ 3x_1 + x_2 + 4x_3 \\ 5x_1 + 7x_2 + 12x_3 \end{bmatrix} = x_1 v_1 + x_2 v_2 + x_3 v_3. \tag{5}$$

Here $v_3 = v_1 + v_2$, so $Ax = (x_1 + x_3)\, v_1 + (x_2 + x_3)\, v_2$.

In general, an $m \times n$ matrix A with columns $a_1, \ldots, a_n$ defines the linear map $T\colon \mathbb{R}^n \to \mathbb{R}^m$, $T(x) = Ax = x_1 a_1 + \cdots + x_n a_n$. Its *image* is the *column space* of A:

$$\mathrm{Im}(A) = \{Ax \mid x \in \mathbb{R}^n\} = \mathrm{span}(a_1, \ldots, a_n).$$

2 Linear Independence and Rank

Definition 19 ▶ Linear Independence

Vectors $v_1, \ldots, v_k$ are **linearly independent** if

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_k = 0.$$

Otherwise they are **linearly dependent**. If $v_1, \ldots, v_k$ are linearly independent, we say that a vector v is *independent of* them if $v, v_1, \ldots, v_k$ are linearly independent.

Definition 20 ▶ Column Rank and Row Rank

The **column rank** $\mathrm{rank}_c(A)$ of a matrix A is the maximum number of linearly independent columns of A. The **row rank** $\mathrm{rank}_r(A)$ is the maximum number of linearly independent rows of A.

Claim 23 (More Vectors Than a Spanning Set)**.** *Let $W = \mathrm{span}(z_1, \ldots, z_r)$. Then any $r+1$ vectors $y_1, \ldots, y_{r+1}$ of W are linearly dependent.*

Proof. A homogeneous system of r linear equations in more than r unknowns has a nonzero solution, by induction on r ($r = 0$: any nonzero vector). If no equation contains c_1, set $c_1 = 1$ and the other unknowns to 0. Otherwise solve one equation for c_1 and substitute into the others: the remaining $r-1$ equations in more than $r-1$ unknowns have a nonzero solution by induction, and the solved equation gives c_1.

Now write $y_j = \sum_{l=1}^{r} t_{lj} z_l$ and take a nonzero solution c of the r equations $\sum_{j=1}^{r+1} t_{lj} c_j = 0$ ($1 \leqslant l \leqslant r$). Then $\sum_j c_j y_j = \sum_l \big(\sum_j t_{lj} c_j\big) z_l = 0$. □

So all bases of W have the same size (a larger one would have more vectors than a set spanning W), the *dimension* $\dim(W)$.

Claim 24 (Column Rank Is the Dimension of the Image)**.** *Let A be an $m \times n$ matrix with $r = \mathrm{rank}_c(A) \geqslant 1$. Then any r linearly independent columns of A form a basis of* $\mathrm{Im}(A)$*. So*

$$\mathrm{rank}_c(A) = \dim(\mathrm{Im}(A)).$$

Proof. Let $w_1, \ldots, w_r$ be independent columns and a any column of A. By the definition of r, the vectors $a, w_1, \ldots, w_r$ are dependent (trivially so if a is one of the w_l): $\beta a + \beta_1 w_1 + \cdots + \beta_r w_r = 0$ with coefficients not all zero. Here $\beta \neq 0$, since otherwise this is a nontrivial relation among the w_l, so $a = -\frac{1}{\beta}(\beta_1 w_1 + \cdots + \beta_r w_r)$. So every column, and hence every combination of columns, lies in $\operatorname{span}(w_1, \ldots, w_r)$: the independent w_l span $\operatorname{Im}(A)$, so they form a basis of it and $\operatorname{rank}_c(A) = r = \dim(\operatorname{Im}(A))$. □

Exercise 12. For the matrix A of Equation (5), find $\operatorname{rank}_c(A)$, $\operatorname{Im}(A)$ and $\operatorname{rank}_r(A)$, and express every row of A as a combination of [1 0 1] and [0 1 1].

Solution. *Columns.* v_1, v_2 are independent: $av_1 + bv_2 = 0$ gives $2a + b = 0$ and $3a + b = 0$, so $a = 0$ and then $b = 0$. The three columns are dependent, since $v_1 + v_2 - v_3 = 0$. So $\operatorname{rank}_c(A) = 2$, and by Claim 24, $\operatorname{Im}(A) = \operatorname{span}(v_1, v_2)$, of dimension 2.
Rows. Let $u_1 = [1\ 0\ 1]$ and $u_2 = [0\ 1\ 1]$, so that $au_1 + bu_2 = [a \quad b \quad a + b]$. Then

$$[2\ 1\ 3] = 2u_1 + 1u_2, \qquad [3\ 1\ 4] = 3u_1 + 1u_2, \qquad [5\ 7\ 12] = 5u_1 + 7u_2.$$

The three rows are dependent, since $16\,[2\ 1\ 3] - 9\,[3\ 1\ 4] - [5\ 7\ 12] = 0$, so $\operatorname{rank}_r(A) \leqslant 2$. The first two rows are independent: $a\,[2\ 1\ 3] + b\,[3\ 1\ 4] = 0$ gives $a + b = 0$ (second entry) and $2a + 3b = 0$ (first entry), so $a = b = 0$. So $\operatorname{rank}_r(A) = 2$. □

Exercise 13. Show that $\operatorname{rank}_r(A) = \operatorname{rank}_c(A)$ for every $m \times n$ matrix A. This common value is called the **rank** of A.

Proof. $\operatorname{rank}_r(A) \leqslant \operatorname{rank}_c(A)$. Let $r = \operatorname{rank}_c(A)$. If $r = 0$, every column of A is zero (a single nonzero vector is independent), so $A = 0$ and $\operatorname{rank}_r(A) = 0$. Let $r \geqslant 1$, let $w_1, \ldots, w_r$ be independent columns of A, and write w_{il} for the ith entry of w_l. By Claim 24, every column a_j of A is $a_j = \sum_{l=1}^{r} s_{lj}\, w_l$ for some numbers s_{lj}. Let $\rho_1, \ldots, \rho_r$ be the rows of the $r \times n$ matrix $S = (s_{lj})$. Entrywise, $a_{ij} = \sum_{l=1}^{r} w_{il}\, s_{lj}$; reading this for fixed i and $j = 1, \ldots, n$,

$$\text{row } i \text{ of } A \;=\; \sum_{l=1}^{r} w_{il}\, \rho_l.$$

So all rows of A lie in $\operatorname{span}(\rho_1, \ldots, \rho_r)$, in which any $r+1$ vectors are dependent (Claim 23): $\operatorname{rank}_r(A) \leqslant r = \operatorname{rank}_c(A)$.
Equality. Applying this to the transpose A^T, whose rows are the columns of A and whose columns are the rows of A, $\operatorname{rank}_c(A) = \operatorname{rank}_r(A^T) \leqslant \operatorname{rank}_c(A^T) = \operatorname{rank}_r(A)$. Together with the first part, equality holds. □

3 Orthogonality

The results of this section hold over $\mathbb{R}$ and over $\mathbb{C}$. The inner product is $\langle a, b\rangle = a \cdot b = \sum_{i=1}^{n} a_i b_i$ for $a, b \in \mathbb{R}^n$, and $\langle a, b\rangle = \sum_{i=1}^{n} \overline{a_i}\, b_i$ (linear in b) for $a, b \in \mathbb{C}^n$. In both cases $\langle a, a\rangle = \sum_{i=1}^{n} |a_i|^2 = \|a\|^2$, which is 0 only for $a = 0$.

Definition 21 ▶ Orthogonality

Two vectors a and b are **orthogonal** if $\langle a, b\rangle = 0$.

Exercise 14. Let $v_1, \dots, v_k$ be linearly independent vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$, and let $v \neq 0$ be orthogonal to each of them. Show that $v, v_1, \dots, v_k$ are linearly independent, so $v \notin \operatorname{span}(v_1, \dots, v_k)$.

Proof. Suppose that $c\,v + c_1 v_1 + \cdots + c_k v_k = 0$. Take the inner product of v with both sides (it is linear in the second argument) and use $\langle v, v_i \rangle = 0$:

$$0 = c\,\langle v, v \rangle + \sum_{i=1}^{k} c_i\,\langle v, v_i \rangle = c\,\|v\|^2.$$

Since $v \neq 0$, $\|v\|^2 > 0$, so $c = 0$. Then $c_1 v_1 + \cdots + c_k v_k = 0$, and the independence of $v_1, \dots, v_k$ gives $c_1 = \cdots = c_k = 0$. If $v = \sum_i \alpha_i v_i$, then $1 \cdot v - \sum_i \alpha_i v_i = 0$ would be a combination with a nonzero coefficient that gives zero, so $v \notin \operatorname{span}(v_1, \dots, v_k)$. □

Week: 5 (September 21–25, 2026)

Lecture Date: Thursday, September 24, 2026

1 Deutsch's Algorithm

Problem 2 ▶ Deutsch's Problem

Given a function $f\colon \{0,1\} \to \{0,1\}$ as a black box, determine whether f is *constant* ($f(0) = f(1)$) or *balanced* ($f(0) \neq f(1)$).

Classically we need 2 queries.

1.1 The Quantum Query U_f

A quantum circuit is reversible, but f need not be. So we use the reversible *quantum version* U_f of f, a controlled-f gate like the CNOT gate of Definition 16 (Figure 11):

$$U_f\colon |a\rangle|b\rangle \longmapsto |a\rangle|b \oplus f(a)\rangle.$$

Applying U_f twice gives back $|a\rangle|b\rangle$, so U_f is reversible. One application of U_f counts as one query to f.

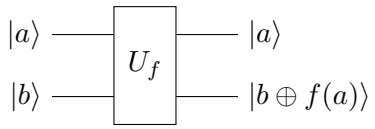


Figure 11: The quantum query U_f.

1.2 The Circuit

Deutsch's algorithm (Figure 12) applies a Hadamard gate to each of $|0\rangle$ and $|1\rangle$, then U_f, then a Hadamard gate to each qubit, and measures.

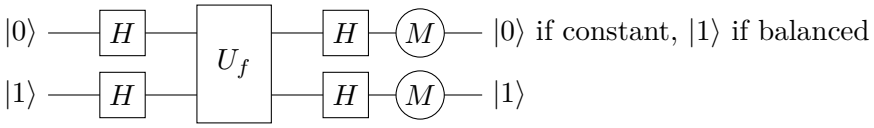


Figure 12: Deutsch's algorithm.

By Equation (3), for $a \in \{0,1\}$,

$$H|a\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle) = \frac{1}{\sqrt{2}} \sum_{b\in\{0,1\}} (-1)^{ab}|b\rangle. \tag{6}$$

Exercise 15. Compute $H\,\frac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle)$ for $a \in \{0,1\}$.

Solution. By linearity and Equation (6) for $H|0\rangle$ and $H|1\rangle$,

$$H\,\tfrac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle) = \tfrac{1}{2}(|0\rangle + |1\rangle) + \tfrac{(-1)^a}{2}(|0\rangle - |1\rangle) = \tfrac{1+(-1)^a}{2}|0\rangle + \tfrac{1-(-1)^a}{2}|1\rangle.$$

For $a = 0$ this is $|0\rangle$ and for $a = 1$ it is $|1\rangle$, so the result is $|a\rangle$. This agrees with $H^2 = I$ (Lemma 20). □

Exercise 16. In the circuit of Figure 12 the second qubit is measured as $|1\rangle$. What does the measurement of the first qubit give? Show that the circuit makes one query to f and that measuring its first qubit gives $f(0) \oplus f(1)$ with probability 1. So the outcome is 0 if f is constant and 1 if f is balanced.

Proof. The circuit contains U_f once, so it makes one query. After the first two Hadamard gates the state is (by Equation (4)) $|\psi_1\rangle = H|0\rangle \otimes H|1\rangle = \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)$. For $c \in \{0,1\}$ we have $|0 \oplus c\rangle - |1 \oplus c\rangle = (-1)^c(|0\rangle - |1\rangle)$ (for $c = 1$ the two kets are swapped). So U_f maps $|a\rangle(|0\rangle - |1\rangle)$ to $(-1)^{f(a)}|a\rangle(|0\rangle - |1\rangle)$. By linearity,

$$|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{\sqrt{2}}\Big((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\Big) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Since $(-1)^{f(1)} = (-1)^{f(0)}(-1)^{f(0)\oplus f(1)}$, the first qubit is $(-1)^{f(0)}\,\frac{1}{\sqrt{2}}\Big(|0\rangle + (-1)^{f(0)\oplus f(1)}|1\rangle\Big)$. Up to the sign $(-1)^{f(0)}$, this is the state of Exercise 15 with $a = f(0) \oplus f(1)$. The second qubit is $H|1\rangle$. So the last two Hadamard gates give $|\psi_3\rangle = (-1)^{f(0)}|f(0) \oplus f(1)\rangle|1\rangle$. The global phase $(-1)^{f(0)}$ does not affect the measurement. The first qubit gives $f(0) \oplus f(1)$ with probability 1, and the second always gives 1. □

2 Phase Flip and the n-Qubit Hadamard Transform

If the second (target) qubit of U_f is in the state $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, then, by the computation in the proof of Exercise 16, for every input x (also for $x \in \{0,1\}^n$, where $U_f\colon |x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$),

$$U_f\,|x\rangle\tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = (-1)^{f(x)}\,|x\rangle\tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

We drop the target qubit and write the query in the *phase flip* form

$$U_f\colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle.$$

By linearity, U_f maps $\sum_x \alpha_x|x\rangle$ to $\sum_x \alpha_x(-1)^{f(x)}|x\rangle$.

2.1 The n-Qubit Hadamard Transform

For $n = 2$ let $|x\rangle = |x_1x_2\rangle = |x_1\rangle|x_2\rangle$. By Equation (4) and then Equation (6) on each qubit,

$$\begin{aligned}(H \otimes H)|x\rangle = H|x_1\rangle \otimes H|x_2\rangle &= \left(\frac{1}{\sqrt{2}}\sum_{y_1\in\{0,1\}}(-1)^{x_1y_1}|y_1\rangle\right)\left(\frac{1}{\sqrt{2}}\sum_{y_2\in\{0,1\}}(-1)^{x_2y_2}|y_2\rangle\right)\\ &= \frac{1}{2}\sum_{y\in\{0,1\}^2}(-1)^{x_1y_1+x_2y_2}|y\rangle = \frac{1}{2}\sum_{y\in\{0,1\}^2}(-1)^{x\cdot y}|y\rangle,\end{aligned}$$

where $x \cdot y = x_1 y_1 \oplus x_2 y_2$, since only the parity of the exponent matters. For example, $(H \otimes H)|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$. In general, for $x, y \in \{0,1\}^n$, the inner product of the bit strings modulo 2 is

$$x \cdot y = x_1 y_1 \oplus x_2 y_2 \oplus \cdots \oplus x_n y_n.$$

Lemma 25 ▶ Walsh–Hadamard Transform

The n-qubit Hadamard transform $H^{\otimes n} = H \otimes H \otimes \cdots \otimes H$ (the *Walsh–Hadamard transform*) acts on a basis state $|x\rangle = |x_1 x_2 \cdots x_n\rangle$ as

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle.$$

Proof. By Equation (4) applied repeatedly, $H^{\otimes n}|x\rangle = H|x_1\rangle \otimes \cdots \otimes H|x_n\rangle$. By Equation (6) the ith factor is $\frac{1}{\sqrt{2}} \sum_{y_i \in \{0,1\}} (-1)^{x_i y_i} |y_i\rangle$. Multiplying out the n sums,

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x_1 y_1 + \cdots + x_n y_n} |y\rangle,$$

and (-1) to the power $x_1 y_1 + \cdots + x_n y_n$ only depends on its parity $x \cdot y$. □

For $x = 0^n = 00\cdots0$ we have $x \cdot y = 0$ for every y, so $H^{\otimes n}|00\cdots0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$, the uniform superposition.

3 Deutsch–Jozsa Algorithm

Problem 3 ▶ Deutsch–Jozsa Problem

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, with the *promise* that f is either constant or balanced (that is, $f(x) = 0$ for exactly half of the 2^n inputs), determine which is the case.

This is a *promise problem*. Let $N = 2^n$ be the number of inputs.

3.1 Classical Query Complexity

Deterministic: in the worst case we need $2^{n-1} + 1 = N/2 + 1$ queries.

Randomized (Monte Carlo): query f on α independent uniformly random inputs and output "balanced" if two answers differ, "constant" otherwise. If f is constant, we are always right. If f is balanced, every answer is 0 or 1 with probability $1/2$ each. We fail only if all α answers agree, which has probability $2 \cdot (1/2)^\alpha = (1/2)^{\alpha-1}$. A constant number of queries gives a constant error probability, and $\alpha = \mathcal{O}(\log N)$ queries give error probability $1/N$.

Quantum: the Deutsch–Jozsa algorithm needs 1 query (and n measurements) and makes no error.

3.2 The Algorithm

The circuit (Figure 13) is Deutsch's circuit with n input qubits in $|0\rangle$. After U_f every qubit gets a second Hadamard gate and is measured. The last qubit always gives 1.

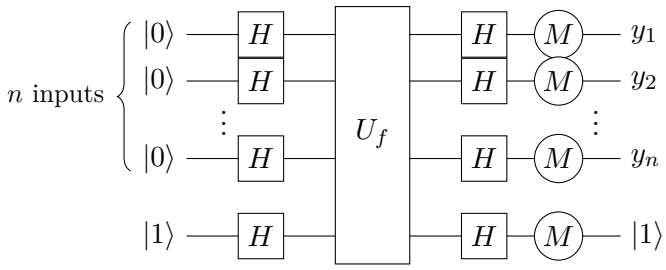


Figure 13: The Deutsch–Jozsa algorithm.

Exercise 17. In the circuit of Figure 13 the state of the n input qubits before the measurement is $\frac{1}{2^n}\sum_y\sum_x(-1)^{f(x)+x\cdot y}|y\rangle$. Show that measuring them gives $y = 0^n$ with probability 1 if f is constant, and with probability 0 if f is balanced. So if any of these n measurements gives 1, then f is balanced; otherwise f is constant.

Proof. We use the phase flip form of U_f. The last qubit is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = H|1\rangle$ before and after the query. Its second Hadamard gate gives $|1\rangle$, so we drop it. The first layer of Hadamard gates maps the input register $|00\cdots0\rangle$ to $\frac{1}{\sqrt{2^n}}\sum_x|x\rangle$. Then the query flips the sign of every $|x\rangle$ with $f(x) = 1$. After the second layer of Hadamard gates, by Lemma 25, the state is

$$\frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}H^{\otimes n}|x\rangle = \frac{1}{2^n}\sum_{y\in\{0,1\}^n}\sum_{x\in\{0,1\}^n}(-1)^{f(x)+x\cdot y}|y\rangle.$$

Since $x\cdot 0^n = 0$ for every x, the amplitude of $|0^n\rangle$ is $\frac{1}{2^n}\sum_x(-1)^{f(x)}$. If f is constant, all 2^n terms are equal and the amplitude is ±1. The amplitudes squared sum to 1, so all others are 0 and the measurement gives 0^n with probability 1. If f is balanced, half the terms are $+1$ and half are -1, so the amplitude is 0. The measurement gives 0^n with probability 0, so at least one of the n measurements gives 1. □

Week: 6 (September 28 – October 2, 2026)

Lecture Date: Tuesday, September 29, 2026

Recap: quantum queries and the Walsh–Hadamard transform. The two algorithms of this lecture use the query $U_f\colon |x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$ (Figure 11; one application is one query) and the *Walsh–Hadamard transform* (Lemma 25) $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle$, where $x \cdot y = x_1y_1 \oplus \cdots \oplus x_ny_n$.

Claim 26 (Sum of Signs)**.** *For every $z \in \{0,1\}^n$,*

$$\sum_{x\in\{0,1\}^n} (-1)^{z\cdot x} = \begin{cases} 2^n & \text{if } z = 0^n, \\ 0 & \text{if } z \neq 0^n. \end{cases}$$

Proof. If $z = 0^n$, all 2^n terms are 1. If $z \neq 0^n$, pick i with $z_i = 1$ and let e_i be the string with a single 1 in position i. The map $x \mapsto x \oplus e_i$ is its own inverse and has no fixed point, so it splits $\{0,1\}^n$ into 2^{n-1} disjoint pairs $\{x, x \oplus e_i\}$. The inner product is linear modulo 2, so $z \cdot (x \oplus e_i) = z \cdot x \oplus z_i = z \cdot x \oplus 1$, and every pair contributes $(-1)^{z\cdot x} + (-1)^{z\cdot x\oplus 1} = 0$. □

In the Deutsch–Jozsa circuit (Figure 13, Exercise 17) the query acts as the phase flip $|k\rangle \mapsto (-1)^{f(k)}|k\rangle$, and the control register after the second layer of Hadamard gates is in the state

$$\sum_{y\in\{0,1\}^n} \alpha_y|y\rangle, \qquad \alpha_y = \frac{1}{2^n} \sum_{k\in\{0,1\}^n} (-1)^{f(k)+k\cdot y}. \tag{7}$$

1 Bernstein–Vazirani Algorithm

Problem 4 ▶ Bernstein–Vazirani Problem

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, with the *promise* that

$$f(x) = s \cdot x = s_1x_1 \oplus s_2x_2 \oplus \cdots \oplus s_nx_n$$

for some unknown n-bit string s, find s.

1.1 The Algorithm

Theorem 27 ▶ Bernstein–Vazirani Algorithm

Run the Deutsch–Jozsa circuit (Figure 13) on $f(x) = s \cdot x$. Then measuring the n control qubits gives $y = s$ with probability 1. So a single query finds s.

Proof. By Equation (7) (with the summation index k renamed x), the control register is in the state $\sum_y \alpha_y|y\rangle$ with

$$\alpha_y = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{s\cdot x\oplus y\cdot x} = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{(s\oplus y)\cdot x},$$

because only the parity of the exponent matters and the inner product is bilinear modulo 2. By Claim 26 with $z = s \oplus y$, $\alpha_y = 1$ if $y = s$ and $\alpha_y = 0$ if $y \neq s$. So the state is $|s\rangle$, and the measurement gives s with probability 1. □

Query complexity versus running time. The query model only counts the calls to U_f. The running time in the *quantum Turing machine* (or quantum circuit) model counts all the gates. The two are not always the same.

2 Simon's Algorithm

Problem 5 ▶ Simon's Problem

Given $f\colon \{0,1\}^n \to \{0,1\}^n$ as a black box, with the *promise* that there is a string $s \in \{0,1\}^n$ such that for all x, y,

$$f(x) = f(y) \iff x = y \text{ or } x = y \oplus s \qquad \text{(equivalently } s = x \oplus y\text{)},$$

find s.

If $s \neq 0^n$, then f is 2-*to*-1 with period s: the two inputs of a pair $\{x, x \oplus s\}$ map to the same output, and different pairs map to different outputs. If $s = 0^n$, then f is 1-*to*-1. So the problem asks whether f is 1-to-1 or 2-to-1 and, in the second case, for the period s. We write $N = 2^n$.

Query complexity. Classically, Simon's problem needs $\Omega\left(\sqrt{N}\right) = \Omega\left(\sqrt{2^n}\right)$ queries, even for randomized algorithms (next lecture). Simon's quantum algorithm uses $\mathcal{O}(n)$ queries and a Gaussian elimination on an $n \times n$ system of linear equations over $\{0,1\}$, which takes $\mathcal{O}(n^3)$ time. This is an exponential speed-up.

2.1 The Circuit

The circuit (Figure 14) has two registers of n qubits each, all starting in $|0\rangle$. Hadamard gates act on the control register before and after the single query U_f. Then the control register is measured, giving a string $y \in \{0,1\}^n$. Because f has n output bits, we use U_f in its original form $|x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$ and not the phase flip form.

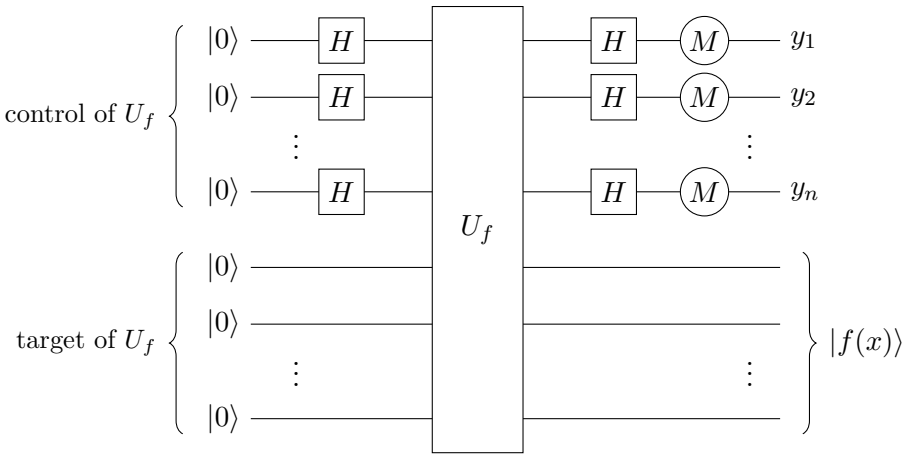


Figure 14: Simon's circuit. The target register is not measured.

2.2 Analysis of One Run

Theorem 28 ▶ One Run of Simon's Circuit

Let y be the string measured by the circuit of Figure 14.

1. If f is 2-to-1 with period $s \neq 0^n$, then y is uniformly distributed over the 2^{n-1} strings y with $s \cdot y = 0$: each of them has probability $1/2^{n-1}$, and every y with $s \cdot y = 1$ has probability 0.
2. If f is 1-to-1, then y is uniformly distributed over all of $\{0,1\}^n$: each string has probability $1/2^n$.

Proof. By Lemma 25, after the first Hadamard gates, the query U_f and the second Hadamard gates the state is

$$\frac{1}{2^n} \sum_{y \in \{0,1\}^n} \sum_{x \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle |f(x)\rangle. \tag{8}$$

Fix y. Its probability is the sum of the squared amplitudes of the states $|y\rangle|z\rangle$.

f is 2-to-1 with period s. Every value z of f has two preimages x_z and $x_z \oplus s$, and $(x_z \oplus s) \cdot y = x_z \cdot y \oplus s \cdot y$. So the amplitude of $|y\rangle|z\rangle$ is

$$\frac{1}{2^n}\Big((-1)^{x_z \cdot y} + (-1)^{(x_z \oplus s) \cdot y}\Big) = \frac{1}{2^n}(-1)^{x_z \cdot y}(1 + (-1)^{s \cdot y}).$$

There are 2^{n-1} values z, so y has probability

$$2^{n-1} \cdot \frac{(1 + (-1)^{s \cdot y})^2}{2^{2n}} = \begin{cases} 1/2^{n-1} & \text{if } s \cdot y = 0, \\ 0 & \text{if } s \cdot y = 1. \end{cases}$$

Exactly 2^{n-1} strings y have $s \cdot y = 0$, by the pairing in the proof of Claim 26.

f is 1-to-1. All $f(x)$ are different, so every state $|y\rangle|f(x)\rangle$ has amplitude $\pm\frac{1}{2^n}$, and y has probability $2^n \cdot \frac{1}{2^{2n}} = \frac{1}{2^n}$. □

2.3 From the Measurements to s

Each run of the circuit gives a random y with $s \cdot y = 0$, which is one linear equation $s_1 y_1 \oplus s_2 y_2 \oplus \cdots \oplus s_n y_n = 0$ for the unknown bits $s_1, \ldots, s_n$. Running the circuit $n-1$ times gives a homogeneous system

$$y_1 \cdot s = 0, \quad y_2 \cdot s = 0, \quad \ldots, \quad y_{n-1} \cdot s = 0$$

of linear equations over the field $\{0,1\}$ (addition is $\oplus$). If $y_1, \ldots, y_{n-1}$ are *linearly independent* over $\{0,1\}$, the system has rank $n-1$. Then its solution space has dimension $n-(n-1)=1$ and consists of 0^n and one nonzero string s'. Gaussian elimination over $\{0,1\}$ tests the independence and computes s' in $\mathcal{O}(n^3)$ time. The true s satisfies every equation. So if $s \neq 0^n$, at most $n-1$ of them can be independent, and $n-1$ independent ones are enough: if the y_i are independent and f is 2-to-1, then $s' = s$.

Theorem 29 ▶ Simon's Algorithm

Algorithm 1 makes $n+1$ queries ($n-1$ quantum, 2 classical) and $\mathcal{O}(n^3)$ other operations. It never outputs a wrong answer, and it outputs the correct s with probability at least $1/4$.

Algorithm 1 Simon's algorithm

run the circuit of Figure 14 $n-1$ times; let $y_1, \ldots, y_{n-1}$ be the measured strings
if $y_1, \ldots, y_{n-1}$ are linearly dependent over $\{0,1\}$: ▷ *Gaussian elimination*
    return FAIL
let s' be the unique nonzero solution of $y_1 \cdot s' = \cdots = y_{n-1} \cdot s' = 0$
if $f(s') = f(0^n)$: ▷ *two classical queries*
    return s' ▷ f *is* 2*-to-*1 *with period* s'
else:
    return 0^n ▷ f *is* 1*-to-*1

Proof. Never wrong. If the algorithm does not fail, $y_1, \ldots, y_{n-1}$ are independent and s' is the unique nonzero solution of the system. If f is 2-to-1 with period s, then s is a nonzero solution, so $s' = s$. The test $f(s') = f(0^n) = f(0^n \oplus s)$ succeeds, and the output $s' = s$ is correct. If f is 1-to-1, then $f(s') \neq f(0^n)$ because $s' \neq 0^n$, and the output $0^n = s$ is correct.

Success probability. The algorithm fails only if the y_i are dependent. The $n-1$ runs are independent (the registers are re-initialized to $|0^n\rangle|0^n\rangle$ for every run). By Theorem 28, the y_i are independent uniform samples from the 2^{n-1} strings with $s \cdot y = 0$ if f is 2-to-1, and from all of $\{0,1\}^n$ if f is 1-to-1. By Exercise 20 (next lecture) they are linearly independent with probability

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{2^3}\right) \cdots \left(1 - \frac{1}{2^{n-1}}\right) \geqslant \frac{1}{4}$$

in the first case and at least $1/2$ in the second. □

2.4 A Deterministic Classical Algorithm

Exercise 18. Find a deterministic classical algorithm for Simon's problem that makes $\mathcal{O}\left(\sqrt{2^n}\right)$ queries.

Solution. Let $m = \lceil n/2 \rceil$ and write $x = (u, v)$, where u is the first m bits of x and v the last $n - m$ bits. Query f on the set

$$S = \{(u, 0^{n-m}) \mid u \in \{0,1\}^m\} \cup \left\{(0^m, v) \mid v \in \{0,1\}^{n-m}\right\},$$

which has at most $2^m + 2^{n-m} \leqslant 2 \cdot 2^{(n+1)/2} = \mathcal{O}\left(\sqrt{2^n}\right)$ elements. If $f(x) = f(x')$ for some $x \neq x'$ in S, output $s = x \oplus x'$. Otherwise output $s = 0^n$.
If $s = (u, v) \neq 0^n$, then $(u, 0^{n-m})$ and $(0^m, v)$ are two different elements of S whose XOR is s, so they collide, and every collision has $x \oplus x' = s$ by the promise. If f is 1-to-1 there is no collision, and the output 0^n is correct. □

Week: 6 (September 28 – October 2, 2026)

Lecture Date: Thursday, October 1, 2026

1 Simon's Problem, Classically

In Simon's problem, $f\colon \{0,1\}^n \to \{0,1\}^n$ is given as a black box with the promise

$$f(x) = f(y) \iff x = y \quad \text{or} \quad x \oplus y = s,$$

and we have to find s (f is 1-to-1 if $s = 0^n$ and 2-to-1 with period s otherwise). Let $N = 2^n$. A classical algorithm learns f only through queries $f(q_1), f(q_2), \ldots$, and we may assume that it never queries the same input twice. A *collision* is a pair of queries $q_i \neq q_j$ with $f(q_i) = f(q_j)$. By the promise a collision gives $s = q_i \oplus q_j$.

1.1 Deterministic Algorithms

Exercise 18 gives a deterministic algorithm with $\mathcal{O}\left(\sqrt{N}\right)$ queries. This is optimal by Corollary 34.

1.2 A Randomized Algorithm

We query random inputs and wait for a collision.

Algorithm 2 Randomized classical algorithm for Simon's problem

choose k distinct inputs $q_1, \ldots, q_k$ uniformly at random (without replacement) and query them
if $f(q_i) = f(q_j)$ for some $i \neq j$:
    return $q_i \oplus q_j$
else:
    return 0^n

Theorem 30 ▶ Randomized Upper Bound

If f is 1-to-1, Algorithm 2 is always correct. If f is 2-to-1 with period s, it is wrong with probability at most $e^{-k(k-1)/(2N)}$. So with $k \geqslant \sqrt{2N \ln N} + 1$ queries (an integer at most N once $n \geqslant 3$) the algorithm is correct with probability at least $1 - 1/N$. With $k \geqslant \sqrt{2N} + 1$ queries, which is $\mathcal{O}\left(\sqrt{N}\right)$, this probability is at least $1 - 1/e$.

Proof. If f is 1-to-1 there is no collision, and the output 0^n is correct. Let f be 2-to-1 with period $s \neq 0^n$. The algorithm is wrong only if no collision occurs. Call $q_j \oplus s$ the *mirror* of q_j. A collision occurs at step i exactly when q_i is the mirror of an earlier query.

Suppose that no collision occurred among $q_1, \ldots, q_{i-1}$. Their $i-1$ mirrors are distinct, and none is among $q_1, \ldots, q_{i-1}$, because that would be a collision. Then all of them are among the $N - (i-1)$ inputs not yet queried, and q_i is uniform among those. So the probability of a collision at step i, given none before, is $(i-1)/(N-i+1) \geqslant (i-1)/N$. Multiplying over the k

steps and using $1 - x \leqslant e^{-x}$,

$$\begin{aligned}\mathbb{P}(\text{no collision among } q_1, \ldots, q_k) &\leqslant \prod_{i=1}^{k}\left(1 - \frac{i-1}{N}\right) \leqslant e^{-\sum_{i=1}^{k}(i-1)/N} \\ &= e^{-k(k-1)/(2N)} \leqslant e^{-(k-1)^2/(2N)}.\end{aligned}$$

With $k - 1 \geqslant \sqrt{2N \ln N}$ this is at most $e^{-\ln N} = 1/N$, and with $k - 1 \geqslant \sqrt{2N}$ it is at most e^{-1}. □

1.3 Lower Bound for Randomized Algorithms via Yao's Principle

Theorem 31 ▶ Randomized Lower Bound

Let $n \geqslant 2$. Every randomized classical algorithm that solves Simon's problem with success probability at least $2/3$ on every input makes more than $\sqrt{N}/2$ queries on some input. So every such algorithm needs $\Omega\left(\sqrt{N}\right) = \Omega\left(\sqrt{2^n}\right)$ queries.

Lemma 32 ▶ Yao's Min-Max Principle, Easy Direction

Let μ be a probability distribution on the inputs. If every deterministic algorithm that makes at most k queries errs with probability greater than δ on an input drawn from μ, then every randomized algorithm that makes at most k queries errs with probability greater than δ on some input.

Proof. A randomized algorithm R is a deterministic algorithm A_r with at most k queries for every fixed random string r. Let $\mathrm{err}(A, x) \in \{0, 1\}$ indicate that A is wrong on input x. Then

$$\max_x \mathbb{P}_r(\mathrm{err}(A_r, x) = 1) \geqslant \mathbb{E}_{x\sim\mu}[\mathbb{P}_r(\mathrm{err}(A_r, x) = 1)] = \mathbb{E}_r[\mathbb{P}_{x\sim\mu}(\mathrm{err}(A_r, x) = 1)] > \delta,$$

since a maximum is at least an average, the two expectations can be exchanged, and every inner probability on the right is greater than δ by assumption. □

The input distribution. Let $\mathcal{D}$ be the following distribution of 2-to-1 functions. Choose s uniformly among the $N - 1$ nonzero strings. This splits $\{0, 1\}^n$ into $N/2$ pairs $\{x, x \oplus s\}$. Assign distinct values to the pairs, uniformly at random among all such assignments.

Principle of deferred decisions. We generate the input lazily. Choose s. When the algorithm queries q_i: if $q_i = q_j \oplus s$ for an earlier query q_j (a collision), answer $f(q_j)$. Otherwise answer a uniformly random value among those not used so far ($N - (i - 1)$ of them as long as no collision has occurred). This produces $\mathcal{D}$, because a uniformly random injective assignment can be built one value at a time, each uniform among the unused ones.

Lemma 33 ▶ Few Queries Rarely Collide

Let $n \geqslant 2$ and let A be a deterministic algorithm that makes $k \leqslant \sqrt{N}/2$ different queries on an input drawn from $\mathcal{D}$. The probability that two of the queries form a pair $\{x, x \oplus s\}$ (a collision) is at most

$$\sum_{i=1}^{k} \frac{i-1}{N - 1 - \binom{i-1}{2}} \leqslant \frac{k(k-1)}{N}.$$

Proof. The ith query q_i collides if $q_i \oplus s = q_j$ for some $j < i$. Suppose there is no collision among $q_1, \ldots, q_{i-1}$, and fix their (distinct) answers. Since A is deterministic, the answers determine $q_1, \ldots, q_i$. No collision rules out the at most $\binom{i-1}{2}$ strings $q_j \oplus q_l$ as values of s. In the lazy generation, these answers have probability $\prod_{j=1}^{i-1} 1/(N-j+1)$ for every other nonzero s, the same for all of them. So, given the answers, s is uniform over the remaining nonzero strings, and there are at least $N - 1 - \binom{i-1}{2}$ of them. The query q_i collides for at most $i-1$ of them (the strings $q_i \oplus q_j$), so

$$\mathbb{P}(q_i \text{ is the first collision}) \leqslant \frac{i-1}{N-1-\binom{i-1}{2}} \leqslant \frac{2(i-1)}{N}.$$

The last step holds because $\binom{i-1}{2} \leqslant k^2/2 \leqslant N/8$ and $N \geqslant 4$, so the denominator is at least $N/2$. Summing over $i = 1, \ldots, k$ gives $\frac{k(k-1)}{N}$. □

Proof of Theorem 31. Let A be a deterministic algorithm with $k \leqslant \sqrt{N}/2$ queries, run on an input drawn from $\mathcal{D}$. We may assume that it never repeats a query. If there is no collision among the k queries, then given the answers the output of A is one fixed string, and s is uniform over at least $N - 1 - \binom{k}{2} \geqslant N/2$ strings (the argument in the proof of Lemma 33, after the kth query). So A outputs s with probability at most $2/N$, and

$$\mathbb{P}(A \text{ correct}) \leqslant \mathbb{P}(\text{collision}) + \frac{2}{N} \leqslant \frac{k(k-1)+2}{N}.$$

For $k \geqslant 2$ this is at most $k^2/N \leqslant 1/4$, and for $k \leqslant 1$ it is $2/N \leqslant 1/2$. So A errs with probability at least $1/2 > 1/3$ on $\mathcal{D}$.

By Lemma 32 with $\mu = \mathcal{D}$ and $\delta = 1/3$, every randomized algorithm with at most $\sqrt{N}/2$ queries errs with probability greater than $1/3$ on some input. So an algorithm with success probability at least $2/3$ on every input makes more than $\sqrt{N}/2$ queries on some input. □

Corollary 34 ▶ Classical Query Complexity of Simon's Problem

The classical query complexity of Simon's problem is $\Theta\left(\sqrt{N}\right) = \Theta\left(\sqrt{2^n}\right)$, both for deterministic algorithms and for randomized algorithms with constant success probability. So Simon's quantum algorithm, with $\mathcal{O}(n)$ queries, is exponentially faster than every classical algorithm.

Proof. A deterministic algorithm that is always correct is a randomized algorithm with success probability $1 \geqslant 2/3$, so Theorem 31 gives the lower bound for both kinds. The upper bounds are Exercise 18 and Theorem 30 (the bound $e^{-(k-1)^2/(2N)}$ in its proof is at most $1/3$ for $k \geqslant \sqrt{2N \ln 3} + 1$). □

Corollary 35 ▶ Decision Version

Let $n \geqslant 2$. Every randomized classical algorithm that decides, with success probability at least $2/3$ on every input, whether f is 1-to-1 or 2-to-1, makes more than $\sqrt{N}/2$ queries on some input.

Proof. Take a uniformly random 1-to-1 function with probability $1/2$ and a function drawn from $\mathcal{D}$ with probability $1/2$. Let A be a deterministic algorithm with at most $\sqrt{N}/2$ queries, all different, and let q be the probability that it answers "1-to-1" on the random 1-to-1 function. In both cases every answer is uniform among the values not used so far, until the first collision. So we can give A the same answers in both cases, and then it gives the same output unless a

collision occurs. A collision has probability at most 1/4 (Lemma 33), so A answers "2-to-1" on $\mathcal{D}$ with probability at most $1 - q + 1/4$. So A is correct with probability at most

$$\frac{1}{2}q + \frac{1}{2}\left(1 - q + \frac{1}{4}\right) = \frac{5}{8},$$

and it errs with probability at least $3/8 > 1/3$. Lemma 32 with $\delta = 1/3$ gives the bound for randomized algorithms. □

Claim 36. *For $a_1, \ldots, a_m \in [0,1]$, $\prod_{i=1}^m (1 - a_i) \geqslant 1 - \sum_{i=1}^m a_i$.*

Proof. We use induction on m, with equality for $m = 1$. Multiplying the hypothesis for $m - 1$ by $1 - a_m \geqslant 0$ gives

$$\prod_{i=1}^{m}(1 - a_i) \geqslant \left(1 - \sum_{i=1}^{m-1} a_i\right)(1 - a_m) = 1 - \sum_{i=1}^{m} a_i + a_m \sum_{i=1}^{m-1} a_i \geqslant 1 - \sum_{i=1}^{m} a_i. \qquad \square$$

2 Simon's Algorithm Revisited

In the state of Equation (8) just before the measurement, the probability of measuring y in the control register is the sum of the squared amplitudes of all basis states $|y\rangle|z\rangle$ with this y. This is the squared norm of the vector that multiplies $|y\rangle$:

$$\mathbb{P}(y) = \left\| \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{x \cdot y} |f(x)\rangle \right\|_2^2.$$

2.1 The 1-to-1 Case

Exercise 19. Suppose that f is 1-to-1. What is the probability of measuring a given string y in Simon's circuit, and do these probabilities add up to 1?

Solution. If f is 1-to-1, the 2^n vectors $|f(x)\rangle$ are distinct basis states, so they are orthonormal. The vector multiplying $|y\rangle$ then has 2^n orthogonal components, each of absolute value $1/2^n$. Its squared norm is

$$\mathbb{P}(y) = \sum_{x \in \{0,1\}^n} \left(\frac{1}{2^n}\right)^2 = 2^n \cdot \frac{1}{2^{2n}} = \frac{1}{2^n}.$$

So the measured y is uniformly distributed over $\{0,1\}^n$, and the probabilities add up to $2^n \cdot 1/2^n = 1$. □

2.2 The 2-to-1 Case

If f is 2-to-1 with period $s \neq 0^n$, every y with $s \cdot y = 0$ is measured with probability $1/2^{n-1}$ and every y with $s \cdot y = 1$ with probability 0 (Theorem 28, case 1). There are 2^{n-1} strings with $s \cdot y = 0$, so the probabilities add up to $2^{n-1} \cdot 1/2^{n-1} = 1$.

2.3 Solving for s

Algorithm 1 runs the circuit $n - 1$ times. In the 2-to-1 case the measured strings $y_1, \ldots, y_{n-1}$ give the system of $n - 1$ linear equations $s \cdot y_1 = 0, \ldots, s \cdot y_{n-1} = 0$ over $\{0,1\}$. If the y_i are linearly independent, the system has rank $n - 1$, and Gaussian elimination gives its only nonzero solution s'. The algorithm then tests $f(0^n) = f(s')$ with two classical queries. If f is 2-to-1,

then s solves the system, so $s' = s$ and the test is true. If f is 1-to-1, the test is false because $s' \neq 0^n$ (Theorem 29).

Exercise 20. Let V be a subspace of $\{0,1\}^n$ of dimension $d \geqslant 1$, so $|V| = 2^d$, and let $y_1, y_2, \ldots$ be independent uniformly random elements of V. Show that

1. $y_1, \ldots, y_d$ are linearly independent with probability

$$\prod_{i=1}^{d}\left(1 - \frac{1}{2^i}\right) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{2^3}\right)\cdots\left(1 - \frac{1}{2^d}\right) \geqslant \frac{1}{4},$$

2. $y_1, \ldots, y_{d-1}$ are linearly independent with probability $\prod_{i=2}^{d}(1 - 1/2^i) \geqslant 1/2$.

Deduce that the strings $y_1, \ldots, y_{n-1}$ of Simon's algorithm are linearly independent with probability at least $1/4$ in the 2-to-1 case ($V = \{y \mid s \cdot y = 0\}$, $d = n - 1$) and at least $1/2$ in the 1-to-1 case ($V = \{0,1\}^n$, $d = n$).

Proof. If $y_1, \ldots, y_j$ are linearly independent, their span has 2^j elements and lies in V, and y_{j+1} is uniform over the 2^d elements of V and independent of $y_1, \ldots, y_j$. So $y_1, \ldots, y_{j+1}$ are linearly independent with probability $1 - 2^j/2^d$. Multiplying for $j = 0, \ldots, m-1$, the first m vectors are linearly independent with probability

$$\prod_{j=0}^{m-1}\left(1 - \frac{1}{2^{d-j}}\right) = \prod_{i=d-m+1}^{d}\left(1 - \frac{1}{2^i}\right),$$

which gives the two products for $m = d$ and $m = d - 1$. By Claim 36,

$$\prod_{i=2}^{d}\left(1 - \frac{1}{2^i}\right) \geqslant 1 - \sum_{i=2}^{d}\frac{1}{2^i} \geqslant \frac{1}{2}, \qquad \prod_{i=1}^{d}\left(1 - \frac{1}{2^i}\right) = \frac{1}{2}\prod_{i=2}^{d}\left(1 - \frac{1}{2^i}\right) \geqslant \frac{1}{4}.$$

In the 2-to-1 case $V = \{y \mid s \cdot y = 0\}$ is the solution set of a linear equation, so it is a subspace, and it has 2^{n-1} elements (Theorem 28). So $d = n - 1$, we have $n - 1 = d$ strings, and part 1 gives $1/4$. In the 1-to-1 case $V = \{0,1\}^n$ (Exercise 19) and $d = n$, we have $d - 1$ strings, and part 2 gives $1/2$. □

Week: 7 (October 5–9, 2026)

Lecture Date: Tuesday, October 6, 2026

Problem 6 ▶ Unstructured Search

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, find an x with $f(x) = 1$.

Think of f as a Boolean formula and of x as an assignment. In *unique satisfiability* we are promised that there is exactly one x with $f(x) = 1$. Let

$$a = |\{x \mid f(x) = 1\}|, \qquad b = |\{x \mid f(x) \neq 1\}| = N - a$$

be the numbers of solutions and of non-solutions. The search is *unstructured*: we know nothing about f except what the queries tell us.

1 Grover's Algorithm on Two Qubits

Let $n = 2$, $N = 4$, and suppose that the unique solution is $x^* = 10$. We write a state of the two qubits as its vector of four amplitudes $(a_{00}, a_{01}, a_{10}, a_{11})$. The algorithm has four steps (Figure 15).

1. *Uniform superposition with Hadamard gates:* $H^{\otimes 2}|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$.

2. *Phase flip.* The query $U_f\colon |x\rangle|c\rangle \mapsto |x\rangle|c \oplus f(x)\rangle$ with the target qubit in $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ is the phase flip of the lecture of September 24, $U_f\colon |x\rangle \mapsto (-1)^{f(x)}|x\rangle$. One query gives $\left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right)$.

3. *Grover diffusion operator.* Let $\mu = \frac{1}{N}\sum_x a_x$ be the mean of the amplitudes. The diffusion operator D reflects every amplitude about the mean (*inversion about the mean*):

$$D\colon a_x \longmapsto 2\mu - a_x \qquad \text{for every } x.$$

   Here $\mu = \frac{1}{4}$, so the three non-solutions become $2 \cdot \frac{1}{4} - \frac{1}{2} = 0$ and the solution becomes $2 \cdot \frac{1}{4} + \frac{1}{2} = 1$. The state is $(0, 0, 1, 0) = |10\rangle$, so measuring gives $x^* = 10$ with probability 1 after a single query.

4. In general, *repeat steps 2 and 3* k times, where k is about $\sqrt{N}$ (Section 3).

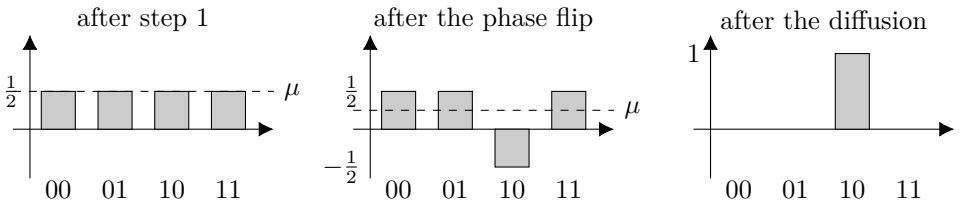


Figure 15: The amplitudes for $N = 4$ and $x^* = 10$ after the Hadamard gates, after the phase flip, and after the inversion about the mean μ.

2 Grover's Algorithm on n Qubits

For n qubits and a unique solution x^* the same steps apply to the N amplitudes. After the Hadamard gates every amplitude is $1/\sqrt{N}$. The phase flip makes the amplitude of x^* equal to $-1/\sqrt{N}$. So the mean is $\mu = (N-2)/(N\sqrt{N}) \approx 1/\sqrt{N}$ and the diffusion operator maps the amplitude of x^* to $2\mu + 1/\sqrt{N} \approx 3/\sqrt{N}$. As long as this amplitude is small, the mean stays close to $1/\sqrt{N}$ and each iteration adds about $2/\sqrt{N}$ to it. So after about $\sqrt{N}$ iterations it is a constant and a measurement finds x^* with constant probability.

2.1 The Operators Z_f, Z_0 and D

The phase flip Z_f. The phase flip

$$Z_f \colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle$$

is implemented by one query U_f with an extra qubit in $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ (Figure 16). The query leaves the extra qubit unchanged, and a second Hadamard gate returns it to $|1\rangle$.

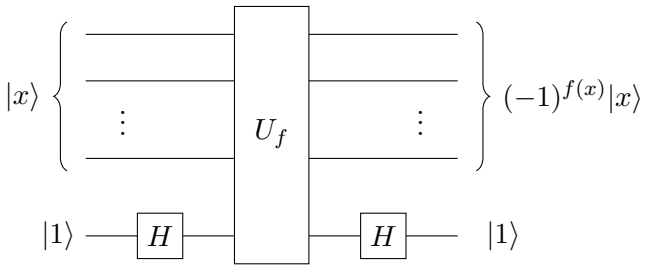


Figure 16: The phase flip Z_f, implemented with one query U_f.

The phase flip Z_0 about 0^n. Let $f_0 \colon \{0,1\}^n \to \{0,1\}$ be the function with $f_0(x) = \bar{x}_1 \wedge \bar{x}_2 \wedge \cdots \wedge \bar{x}_n$, where $\bar{x}_i = \neg x_i$, so $f_0(x) = 1$ only for $x = 0^n$. Its phase flip is

$$Z_0 \colon |x\rangle \longmapsto \begin{cases} -|x\rangle & \text{if } x = 0^n, \\ |x\rangle & \text{otherwise,} \end{cases} \qquad \text{so} \qquad Z_0 = I - 2|0^n\rangle\langle 0^n|.$$

As a matrix, Z_0 is the $N \times N$ diagonal matrix with -1 in the top left corner and 1 elsewhere.

The diffusion operator D. Let $|h\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$ be the uniform superposition. The diffusion operator is

$$D = -H^{\otimes n} Z_0 H^{\otimes n}.$$

Lemma 37 ▶ Inversion About the Mean

$D = 2|h\rangle\langle h| - I$, and for every state $\sum_x a_x|x\rangle$ with mean amplitude $\mu = \frac{1}{N}\sum_x a_x$,

$$D \sum_{x\in\{0,1\}^n} a_x|x\rangle = \sum_{x\in\{0,1\}^n} (2\mu - a_x)|x\rangle.$$

Proof. Since $H^\dagger = H$ and $H^2 = I$, we have $H^{\otimes n}H^{\otimes n} = I$ and $\langle 0^n|H^{\otimes n} = (H^{\otimes n}|0^n\rangle)^\dagger = \langle h|$, so

$$D = -H^{\otimes n}(I - 2|0^n\rangle\langle 0^n|)H^{\otimes n} = -(I - 2|h\rangle\langle h|) = 2|h\rangle\langle h| - I.$$

For $|\psi\rangle = \sum_x a_x|x\rangle$ we have $\langle h|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x a_x = \sqrt{N}\mu$, so

$$D|\psi\rangle = 2\sqrt{N}\mu|h\rangle - |\psi\rangle = 2\mu\sum_x |x\rangle - \sum_x a_x|x\rangle = \sum_x (2\mu - a_x)|x\rangle. \qquad \square$$

So D is the inversion about the mean of step 3.

The Grover iteration. One iteration of steps 2 and 3 is the operator

$$G = DZ_f = -H^{\otimes n} Z_0 H^{\otimes n} Z_f.$$

Grover's algorithm prepares $|h\rangle = H^{\otimes n}|0^n\rangle$, applies G k times and measures (Figure 17).

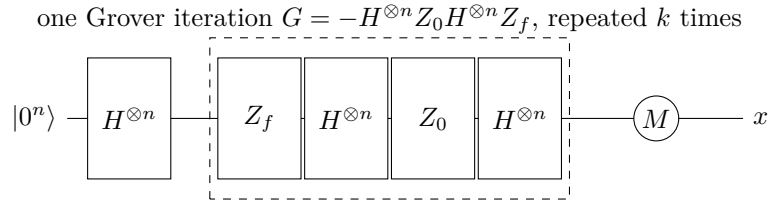


Figure 17: Grover's algorithm on n qubits (the extra qubit used by Z_f and Z_0 is not drawn).

3 Analysis: A Rotation in Two Dimensions

Let f have a solutions, $1 \leqslant a \leqslant N-1$, and let $b = N - a$. Let

$$|A\rangle = \frac{1}{\sqrt{a}} \sum_{\substack{x\in\{0,1\}^n \\ f(x)=1}} |x\rangle, \qquad |B\rangle = \frac{1}{\sqrt{b}} \sum_{\substack{x\in\{0,1\}^n \\ f(x)=0}} |x\rangle$$

be the uniform superpositions of the solutions and of the non-solutions. They are unit vectors, and they are orthogonal since no basis state appears in both. The whole computation stays in the plane they span. The initial state is

$$|h\rangle = \frac{1}{\sqrt{N}} \sum_{x\in\{0,1\}^n} |x\rangle = \frac{1}{\sqrt{N}}\Big(\sqrt{a}|A\rangle + \sqrt{b}|B\rangle\Big) = \sqrt{\frac{a}{N}}|A\rangle + \sqrt{\frac{b}{N}}|B\rangle. \tag{9}$$

Since $\frac{a}{N} + \frac{b}{N} = 1$, there is a unique angle $\theta \in (0, \pi/2)$ with

$$\sin\theta = \sqrt{\frac{a}{N}}, \qquad \cos\theta = \sqrt{\frac{b}{N}}, \qquad \text{so that} \qquad |h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle.$$

For a unique solution, $\sin\theta = 1/\sqrt{N}$, so $\theta \approx 1/\sqrt{N}$ and $|h\rangle$ is almost $|B\rangle$ (Figure 18).

Lemma 38 ▶ One Grover Iteration

$$G|A\rangle = \left(1 - \frac{2a}{N}\right)|A\rangle - \frac{2\sqrt{ab}}{N}|B\rangle = \cos 2\theta|A\rangle - \sin 2\theta|B\rangle,$$
$$G|B\rangle = \frac{2\sqrt{ab}}{N}|A\rangle - \left(1 - \frac{2b}{N}\right)|B\rangle = \sin 2\theta|A\rangle + \cos 2\theta|B\rangle.$$

Proof. By the definition of Z_f, $Z_f|A\rangle = -|A\rangle$ and $Z_f|B\rangle = |B\rangle$. By Equation (9) and orthonormality, $\langle h|A\rangle = \sqrt{a/N}$ and $\langle h|B\rangle = \sqrt{b/N}$. With $D = 2|h\rangle\langle h| - I$ (Lemma 37), substituting

Equation (9) for $|h\rangle$ in the last step of each line,

$$G|A\rangle = -D|A\rangle = |A\rangle - 2|h\rangle\langle h|A\rangle = |A\rangle - 2\sqrt{\frac{a}{N}}|h\rangle = \left(1 - \frac{2a}{N}\right)|A\rangle - \frac{2\sqrt{ab}}{N}|B\rangle,$$
$$G|B\rangle = D|B\rangle = 2|h\rangle\langle h|B\rangle - |B\rangle = 2\sqrt{\frac{b}{N}}|h\rangle - |B\rangle = \frac{2\sqrt{ab}}{N}|A\rangle - \left(1 - \frac{2b}{N}\right)|B\rangle.$$

The trigonometric form follows from $\sin^2\theta = a/N$, $\cos^2\theta = b/N$ and the double angle formulas: $\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2a/N$, $\cos 2\theta = 2\cos^2\theta - 1 = -(1 - 2b/N)$ and $\sin 2\theta = 2\sin\theta\cos\theta = 2\sqrt{ab}/N$. □

Since $a + b = N$, both $1 - 2a/N$ and $2b/N - 1$ equal $(b-a)/N$. We write a state $\beta|B\rangle + \alpha|A\rangle$ of the plane as the column vector $(\beta, \alpha)^T$ in the basis $(|B\rangle, |A\rangle)$. Then G acts as the 2×2 matrix

$$G = \begin{pmatrix} \frac{b-a}{N} & -\frac{2\sqrt{ab}}{N} \\ \frac{2\sqrt{ab}}{N} & \frac{b-a}{N} \end{pmatrix} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \tag{10}$$

whose columns are the coordinates of $G|B\rangle$ and $G|A\rangle$. This is the rotation matrix $R_\phi = \left(\begin{smallmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{smallmatrix}\right)$ with $\phi = 2\theta$. So one Grover iteration *rotates* the plane by the angle 2θ, from $|B\rangle$ towards $|A\rangle$.

Exercise 21. Compute $G|h\rangle = \sqrt{a/N}\,G|A\rangle + \sqrt{b/N}\,G|B\rangle$ and $G^k|h\rangle$ for $k \geqslant 0$, and verify that the matrix of G in Equation (10) is the rotation $R_{2\theta}$.

Solution. *The matrix is a rotation.* By Lemma 38 and $(b-a)/N = 1 - 2a/N = 2b/N - 1$, the two columns of the matrix in Equation (10) are the coordinates of $G|B\rangle = \cos 2\theta|B\rangle + \sin 2\theta|A\rangle$ and $G|A\rangle = -\sin 2\theta|B\rangle + \cos 2\theta|A\rangle$, which are the columns of $R_{2\theta}$ (not R_θ).
The state $G|h\rangle$. By Equation (9), $|h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle$. By Lemma 38 and the addition formulas for $\sin(\theta + 2\theta)$ and $\cos(\theta + 2\theta)$,

$$\begin{aligned} G|h\rangle &= \sin\theta(\cos 2\theta|A\rangle - \sin 2\theta|B\rangle) + \cos\theta(\sin 2\theta|A\rangle + \cos 2\theta|B\rangle) \\ &= \sin 3\theta|A\rangle + \cos 3\theta|B\rangle. \end{aligned}$$

With $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $4b/N - 3 = 1 - 4a/N$, this is

$$G|h\rangle = \sqrt{\frac{a}{N}}\left(3 - \frac{4a}{N}\right)|A\rangle + \sqrt{\frac{b}{N}}\left(1 - \frac{4a}{N}\right)|B\rangle.$$

The state starts at the angle θ from $|B\rangle$ and the iteration adds 2θ (Figure 18).
The state $G^k|h\rangle$. The same computation with any angle β in place of θ gives $G(\sin\beta|A\rangle + \cos\beta|B\rangle) = \sin(\beta + 2\theta)|A\rangle + \cos(\beta + 2\theta)|B\rangle$. So by induction on k, starting from the angle θ of $|h\rangle$,

$$G^k|h\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

□

The number of iterations is computed in the next lecture (Section 2).

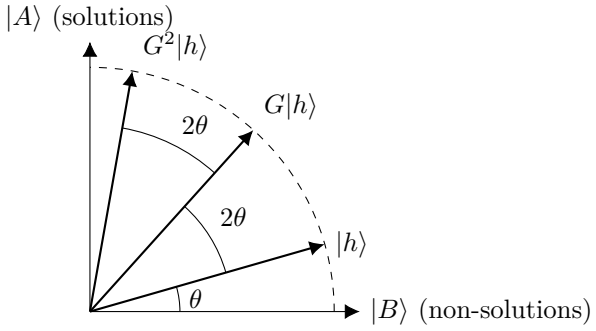


Figure 18: Grover's algorithm in the plane spanned by $|A\rangle$ and $|B\rangle$ for $\theta = 16°$.

Week: 7 (October 5–9, 2026)

Lecture Date: Thursday, October 8, 2026

Recap. We keep the notation of the last lecture (a, $b = N - a$ with $1 \leqslant a \leqslant N-1$, $|A\rangle$, $|B\rangle$, Z_f, Z_0, D, $|h\rangle$ and the Grover iteration $G = DZ_f = -H^{\otimes n} Z_0 H^{\otimes n} Z_f$; Figure 17). By Equation (9), $|h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle$ with $\theta = \sin^{-1}\left(\sqrt{a/N}\right) \in (0, \pi/2)$. We write a unit vector of the plane spanned by $|A\rangle$ and $|B\rangle$ with real coefficients as $\sin\alpha|A\rangle + \cos\alpha|B\rangle$ and call α its *angle* (measured from $|B\rangle$ towards $|A\rangle$). So $|h\rangle$ has angle θ, and a measurement of the state with angle α gives a solution with probability $\sin^2\alpha$.

1 A Grover Iteration Is Two Reflections

Since $Z_f|A\rangle = -|A\rangle$ and $Z_f|B\rangle = |B\rangle$ (proof of Lemma 38), $Z_f(\sin\alpha|A\rangle + \cos\alpha|B\rangle) = \sin(-\alpha)|A\rangle + \cos(-\alpha)|B\rangle$. So Z_f is the reflection of the plane about the line of $|B\rangle$.

Let $|h^\perp\rangle = \cos\theta|A\rangle - \sin\theta|B\rangle$. It is a unit vector of the plane with $\langle h|h^\perp\rangle = \sin\theta\cos\theta - \cos\theta\sin\theta = 0$, so $|h\rangle$ and $|h^\perp\rangle$ are a second orthonormal basis of the plane (Figure 19).

Exercise 22. Show that $D = 2|h\rangle\langle h| - I$, and that for every state $|\phi\rangle = c|h\rangle + d|h^\perp\rangle$ of the plane,

$$D|\phi\rangle = c|h\rangle - d|h^\perp\rangle.$$

Solution. The first claim is Lemma 37: $H^{\otimes n}H^{\otimes n} = I$ and $\langle 0^n|H^{\otimes n} = \langle h|$ give

$$D = -H^{\otimes n}(I - 2|0^n\rangle\langle 0^n|)H^{\otimes n} = 2|h\rangle\langle h| - I.$$

Since $\langle h|h\rangle = 1$ and $\langle h|h^\perp\rangle = 0$, we have $\langle h|\phi\rangle = c$, so

$$D|\phi\rangle = 2|h\rangle\langle h|\phi\rangle - |\phi\rangle = 2c|h\rangle - c|h\rangle - d|h^\perp\rangle = c|h\rangle - d|h^\perp\rangle.$$

So D is the reflection of the plane about the line of $|h\rangle$. □

1.1 Two Reflections Make a Rotation

In Figure 19 the first reflection takes the angle θ to $-\theta$ and the second, about the line at angle θ, takes $-\theta$ to 3θ.

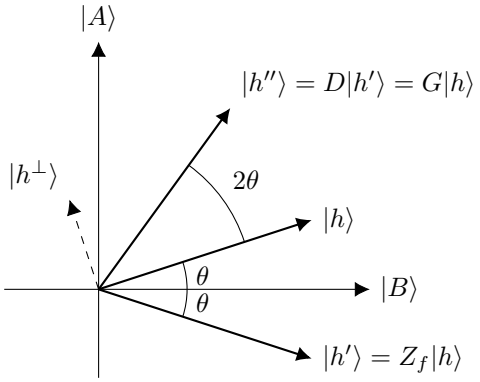


Figure 19: One Grover iteration applied to $|h\rangle$: Z_f reflects about the line of $|B\rangle$, then D reflects about the line of $|h\rangle$.

Claim 39 (Two Reflections Make a Rotation)**.** *Let $|A\rangle$ and $|B\rangle$ be orthonormal vectors, let θ be a real number and $|\psi\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle$. Let S be a linear operator with $S|A\rangle = -|A\rangle$ and $S|B\rangle = |B\rangle$, and let $T = 2|\psi\rangle\langle\psi| - I$. Then for every real number β,*

$$TS(\sin\beta|A\rangle + \cos\beta|B\rangle) = \sin(\beta + 2\theta)|A\rangle + \cos(\beta + 2\theta)|B\rangle,$$

and so, for every integer $k \geqslant 0$,

$$(TS)^k|\psi\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

Proof. By linearity, $S(\sin\beta|A\rangle + \cos\beta|B\rangle) = -\sin\beta|A\rangle + \cos\beta|B\rangle = \sin\gamma|A\rangle + \cos\gamma|B\rangle =: |\chi\rangle$ with $\gamma = -\beta$. By orthonormality, $\langle\psi|\chi\rangle = \sin\theta\sin\gamma + \cos\theta\cos\gamma = \cos(\gamma - \theta)$, so

$$\begin{aligned} T|\chi\rangle &= 2\cos(\gamma-\theta)|\psi\rangle - |\chi\rangle \\ &= (2\cos(\gamma-\theta)\sin\theta - \sin\gamma)|A\rangle + (2\cos(\gamma-\theta)\cos\theta - \cos\gamma)|B\rangle. \end{aligned}$$

By the product formulas $2\cos u\sin v = \sin(u+v) - \sin(u-v)$ and $2\cos u\cos v = \cos(u+v) + \cos(u-v)$ with $u = \gamma - \theta$ and $v = \theta$, the coefficients are $-\sin(\gamma - 2\theta) = \sin(2\theta - \gamma)$ and $\cos(\gamma - 2\theta) = \cos(2\theta - \gamma)$. So T sends the angle γ to $2\theta - \gamma$, and TS sends β to $2\theta - (-\beta) = \beta + 2\theta$. This is the first claim. The state $|\psi\rangle$ has the angle θ and each application of TS adds 2θ, so induction on k gives the angle $(2k+1)\theta$ for $(TS)^k|\psi\rangle$. □

For Grover's algorithm we take $S = Z_f$, $|\psi\rangle = |h\rangle$ and $T = D = 2|h\rangle\langle h| - I$ (Exercise 22), so that $TS = DZ_f = G$:

Corollary 40 ▶ The State After k Iterations

Every Grover iteration rotates the plane by 2θ from $|B\rangle$ towards $|A\rangle$, and the state after k iterations is

$$|h^k\rangle = G^k|h\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

A measurement of $|h^k\rangle$ gives a solution with probability $\sin^2((2k+1)\theta)$.

1.2 The Matrix of a Grover Iteration

Let M be the matrix of the Grover iteration in the basis $(|B\rangle, |A\rangle)$ (Equation (10)).

Exercise 23. Show that

$$M = \begin{pmatrix} \sqrt{b/N} & -\sqrt{a/N} \\ \sqrt{a/N} & \sqrt{b/N} \end{pmatrix}^2 = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^2,$$

that is, the Grover iteration is the rotation by θ applied twice.

Solution. Let $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ with $\cos\theta = \sqrt{b/N}$ and $\sin\theta = \sqrt{a/N}$. Multiplying,

$$R_\theta^2 = \begin{pmatrix} \cos^2\theta - \sin^2\theta & -2\sin\theta\cos\theta \\ 2\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{pmatrix} = \begin{pmatrix} \dfrac{b-a}{N} & -\dfrac{2\sqrt{ab}}{N} \\ \dfrac{2\sqrt{ab}}{N} & \dfrac{b-a}{N} \end{pmatrix} = M,$$

and by the double angle formulas the middle matrix is $R_{2\theta}$. So $M^k = R_{2k\theta}$. This is Corollary 40 in matrix form: $|h\rangle$ is the first column $(\cos\theta, \sin\theta)^T$ of R_θ, so $M^k|h\rangle$ is the first column of $R_{(2k+1)\theta}$. □

2 The Number of Iterations

By Corollary 40 we want $(2k+1)\theta \approx \pi/2$, so we take k to be an integer closest to $\frac{\pi}{4\theta} - \frac{1}{2}$. Since $\theta \approx \sin\theta = \sqrt{a/N}$ for small a/N, this is

$$k \approx \frac{\pi}{4}\sqrt{\frac{N}{a}}, \qquad \text{for a unique solution} \quad k \approx \frac{\pi}{4}\sqrt{N}.$$

Exercise 24. Let f have a solutions, $1 \leqslant a \leqslant N-1$, and let k be an integer closest to $\frac{\pi}{4\theta} - \frac{1}{2}$. Show that Grover's algorithm with k iterations outputs a solution with probability at least $1 - a/N$.

Proof. Let $t = \frac{\pi}{4\theta} - \frac{1}{2}$, so that $(2t+1)\theta = \pi/2$. Then $|k - t| \leqslant 1/2$ and

$$(2k+1)\theta = \frac{\pi}{2} + 2(k-t)\theta, \qquad |2(k-t)\theta| \leqslant \theta.$$

By Corollary 40 and $\sin(\pi/2 + x) = \cos x$, the success probability is

$$\sin^2((2k+1)\theta) = \cos^2(2(k-t)\theta) \geqslant \cos^2\theta = 1 - \frac{a}{N},$$

because the cosine is even and decreases on $[0, \pi/2]$, and $\sin^2\theta = a/N$. □

3 A Random Number of Iterations

The number of iterations, about $\frac{\pi}{4}\sqrt{N/a}$, depends on the number a of solutions, which we may not know. So we choose k *at random*. Since $\sqrt{N}$ need not be an integer, the largest value is $\lfloor 2\sqrt{N} \rfloor$.

Exercise 25. Let $N = 2^n$, let f have a solutions, $1 \leqslant a \leqslant N/2$, and let $K = \lfloor 2\sqrt{N} \rfloor$. Choose k uniformly at random from $\{1, \ldots, K\}$, make k Grover iterations on $|h\rangle$ and

measure. Prove that the probability of measuring a solution is

$$\mathbb{P}(\text{success}) = \frac{1}{2} - \frac{\sin(4(K+1)\theta) - \sin 4\theta}{4K \sin 2\theta} \geqslant \frac{1}{2} - \frac{1}{4K \sin 2\theta} \geqslant \frac{3}{8}, \tag{11}$$

so that $\mathbb{P}(\text{success}) = \Theta(1)$ with at most $2\sqrt{N} = \mathcal{O}\left(\sqrt{N}\right)$ queries.

Proof. The average. Given k, the success probability is $\sin^2((2k+1)\theta)$ by Corollary 40, and each value has probability $1/K$. By the law of total probability and $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

$$\mathbb{P}(\text{success}) = \frac{1}{K}\sum_{k=1}^{K} \sin^2((2k+1)\theta) = \frac{1}{2} - \frac{1}{2K}\sum_{k=1}^{K} \cos((4k+2)\theta).$$

By the product formula $2\cos u \sin v = \sin(u+v) - \sin(u-v)$ with $u = (4k+2)\theta$ and $v = 2\theta$, the sum telescopes:

$$2\sin 2\theta \sum_{k=1}^{K} \cos((4k+2)\theta) = \sum_{k=1}^{K} (\sin(4(k+1)\theta) - \sin(4k\theta)) = \sin(4(K+1)\theta) - \sin 4\theta.$$

As $\sin\theta = \sqrt{a/N} \leqslant 1/\sqrt{2}$, we have $0 < \theta \leqslant \pi/4$ and $\sin 2\theta > 0$, so dividing by $2\sin 2\theta$ gives the equality in Equation (11).
The lower bound. $\sin(4(K+1)\theta) \leqslant 1$, and $\sin 4\theta \geqslant 0$ because $0 < 4\theta \leqslant \pi$. So the numerator is at most 1, which gives the first inequality in Equation (11). Since $b = N - a \geqslant N/2$,

$$\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\sqrt{ab}}{N} \geqslant \frac{2\sqrt{a \cdot N/2}}{N} = \sqrt{\frac{2a}{N}}.$$

Next, $\lfloor 2\sqrt{N} \rfloor \geqslant \sqrt{2N}$ for $N = 2^n$. If n is even, $2\sqrt{N}$ is an integer. If n is odd, $\sqrt{2N} = 2^{(n+1)/2}$ is an integer with $\sqrt{2N} \leqslant 2\sqrt{N}$, so it is at most the floor. So $K\sin 2\theta \geqslant \sqrt{2N} \cdot \sqrt{2a/N} = 2\sqrt{a} \geqslant 2$ and $\mathbb{P}(\text{success}) \geqslant \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$. A probability is at most 1, so $\mathbb{P}(\text{success}) = \Theta(1)$. The number of queries is $k \leqslant K \leqslant 2\sqrt{N}$. □

For the set $\left\{1, \ldots, \lfloor\sqrt{N}\rfloor\right\}$ the same proof with $K = \lfloor\sqrt{N}\rfloor \geqslant \sqrt{N/2}$ gives $\mathbb{P}(\text{success}) \geqslant 1/4$.

4 Quantum Amplitude Amplification

Let $\mathcal{A}$ be any quantum algorithm without measurements (a unitary operator on n qubits), and let $f\colon \{0,1\}^n \to \{0,1\}$ tell us which outcomes x are *good* ($f(x) = 1$) and which are *bad* ($f(x) = 0$). Write $\mathcal{A}|0^n\rangle = \sum_x c_x|x\rangle$, let $p = \sum_{x|f(x)=1} |c_x|^2$ be the probability that a measurement of $\mathcal{A}|0^n\rangle$ gives a good outcome, and assume $0 < p < 1$. With the unit vectors

$$|G\rangle = \frac{1}{\sqrt{p}} \sum_{x|f(x)=1} c_x|x\rangle, \qquad |B\rangle = \frac{1}{\sqrt{1-p}} \sum_{x|f(x)=0} c_x|x\rangle,$$

which are orthogonal because no basis state appears in both, the state that $\mathcal{A}$ produces is $|\phi\rangle = \mathcal{A}|0^n\rangle = \sqrt{p}\,|G\rangle + \sqrt{1-p}\,|B\rangle$. Here $|G\rangle$ is the good state, unrelated to the Grover iteration G. *Classically* we would run $\mathcal{A}$, measure, check the outcome with f, and repeat until the outcome is good. The number of runs is geometric with mean $1/p$.

Exercise 26. Let $\theta \in (0, \pi/2)$ with $\sin\theta = \sqrt{p}$, and let

$$Q = -\mathcal{A}Z_0\mathcal{A}^{-1}Z_f.$$

Show that $Q^k|\phi\rangle = \sin((2k+1)\theta)|G\rangle + \cos((2k+1)\theta)|B\rangle$ for every $k \geqslant 0$. Show that for $k = \lfloor \pi/4\theta \rfloor$, a measurement of $Q^k|\phi\rangle$ gives a good outcome with probability at least $\max(p, 1-p) \geqslant 1/2$, and that computing $Q^k\mathcal{A}|0^n\rangle$ takes $2k+1 \leqslant \frac{\pi}{2\sqrt{p}} + 1 = \mathcal{O}(1/\sqrt{p})$ applications of $\mathcal{A}$ or $\mathcal{A}^{-1}$ and k queries to f.

Proof. $Z_f|G\rangle = -|G\rangle$ and $Z_f|B\rangle = |B\rangle$. Since $\mathcal{A}$ is unitary, $\langle 0^n|\mathcal{A}^{-1} = (\mathcal{A}|0^n\rangle)^\dagger = \langle\phi|$, so

$$-\mathcal{A}Z_0\mathcal{A}^{-1} = -\mathcal{A}(I - 2|0^n\rangle\langle 0^n|)\mathcal{A}^{-1} = 2|\phi\rangle\langle\phi| - I.$$

So Q is the reflection about $|B\rangle$ followed by the reflection about $|\phi\rangle = \sin\theta|G\rangle + \cos\theta|B\rangle$, and Claim 39 gives the formula for $Q^k|\phi\rangle$.

$|G\rangle$ contains only good basis states and $|B\rangle$ only bad ones, so the success probability is $\sin^2((2k+1)\theta)$. For $k = \lfloor \pi/4\theta \rfloor$ we have $\frac{\pi}{2} - \theta < (2k+1)\theta \leqslant \frac{\pi}{2} + \theta$, so it is at least $\cos^2\theta = 1 - p$. If $p \leqslant 1/2$, then $1 - p = \max(p, 1-p)$. If $p > 1/2$, then $\theta > \pi/4$, so $k = 0$ and it is $\sin^2\theta = p$.

Each application of Q uses $\mathcal{A}$ once, $\mathcal{A}^{-1}$ once and one query to f, and preparing $|\phi\rangle$ uses $\mathcal{A}$ once more. So the cost is $2k+1$ applications and k queries. Since $\sin\theta \leqslant \theta$, $k \leqslant \frac{\pi}{4\theta} \leqslant \frac{\pi}{4\sqrt{p}}$, so $2k+1 \leqslant \frac{\pi}{2\sqrt{p}} + 1$. □

AI Use Declaration

Tool: Claude Code (Anthropic).

I used AI to transcribe my handwritten notes from class and convert them to LaTeX. I used AI as a verifier and to explore: I worked out the steps and the ways to solve the problems manually, and then used AI to verify them. The figures and examples come from the partial examples done in class, and some are AI generated for practice.